

Highlights

TrustFedGNN: Contribution-Based Aggregation without Self-Reported Metadata for Trustworthy Federated Graph Learning

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- Client-declared fairness metadata captures six published FL aggregation rules.
- Falsified reports raise disparity up to 7x and cut AUC by 0.14 over 30 seeds.
- Multi-round client-wise DP drives constraint testing toward chance.
- Server-referenced aggregation is immune to the channel by construction.

TrustFedGNN: Contribution-Based Aggregation without Self-Reported Metadata for Trustworthy Federated Graph Learning

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Abstract

Cross-silo federated graph learning for credit scoring, bail assessment and anti-money-laundering must satisfy fairness, privacy and Byzantine resilience at once. Every fairness-aware aggregator we surveyed obtains its fairness signal by asking clients to declare it, making the declaration an attack surface. Attacking six published aggregation rules with the poisoned update held fixed, isolating the declared fields, falsification raised the adversary’s weight in 141 of 144 non-tied seed pairs over thirty seeds; two rules reading no declared field are bit-exact. Privatising the report does not repair it: we prove that once noise dominates, the quantity the server observes converges to a constant of the noise, independent of every true disparity, so the budget test measures the noise multiplier rather than fairness. The obstruction is conditional—it bites in small cross-silo deployments over many rounds, not in large cross-device settings with one release. On German the reported aggregate departs from the true one by over 700% of the budget at $\epsilon = 8$, and the exact folded-normal prediction matches measurement at every privacy level tested. TrustFedGNN removes the channel instead of hardening it: the server builds a bi-objective target gradient on a client-inaccessible holdout and scores clients by alignment with it, so no declared field enters the weights and immunity is syntactic, not empirical. The price: fairness cannot be steered by asking, the server needs an untouched holdout, and on

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clean data a Byzantine defence with no fairness mechanism attains higher AUC than we do, with weights oscillating 7.4 times more. Fairness-Targeted Gradient Decomposition separately certifies (ϵ, δ) -DP for the two-dimensional group-mean statistic, not the update, at $O(1)$ noise and 0.00175 AUC against 0.20163 for full DP-SGD. Four algebraic properties are machine-checked in Lean 4, and we report seven pre-registered hypotheses our own measurements refuted.

Keywords: Federated Learning, Graph Neural Networks, Algorithmic Fairness, Differential Privacy, Byzantine Robustness, Contribution Valuation

1. Introduction

Graph neural networks (GNNs) are now the default model class for fraud detection, anti-money-laundering (AML) and credit assessment [1, 2, 3], because the entities in those domains are defined as much by their relations—transactions, shared devices, social ties—as by their own attributes. In cross-institutional settings, however, data-protection regimes such as the EU General Data Protection Regulation forbid pooling those graphs into one repository. The resulting setting is *cross-silo horizontal federated graph learning*: K institutions each hold an induced subgraph, inter-silo edges are severed, and only model updates cross the boundary [4, 5].

Deploying such a system in a high-stakes domain requires three properties at once, and each pair of them is in tension.

Structural bias against group fairness.. Nodes sharing a sensitive attribute are densely interconnected—demographic homophily—so recursive neighbourhood aggregation propagates and amplifies historical bias [6, 7].

Group fairness against differential privacy.. Enforcing a fairness constraint requires the server to observe some disparity signal, and that signal leaks group membership. The textbook remedy, DP-SGD over the full parameter vector [8], injects noise scaling as $O(\sqrt{P})$ and destroys GNN utility: in our own ablation it costs 0.2016 AUC (Section 7.2).

Fairness steering against Byzantine resilience.. This is the tension we consider unresolved, and it motivates the paper. Fairness-aware federated aggregators—FairFed [9], PUFFLE [10], F²GNN [11], FairGFL [12]—compute aggregation

weights from disparity statistics *that each client reports about itself*. Nothing in those protocols verifies the report. An adversary may therefore transmit a poisoned update while declaring $\widehat{\text{DPD}}_k = 0$, and the fairness mechanism will reward it precisely because it claims to be fair. Kasyap et al. [13] and EAB-FL [14] demonstrate this class of attack; classical Byzantine defences [15, 16, 17] do not address it, because they screen the parameter update and never examine the metadata channel at all. We measure how general the weakness is by attacking six published aggregation rules end to end, holding the poisoned update fixed and varying only the accompanying report: lying raised the adversary’s aggregation weight in 141 of 144 non-tied seed pairs and lowered it in three, and two rules that read no declared field are bit-exact under the same falsification (Section 7.1).

The obvious repair—keep the reported statistic but privatise it—does not work either, and the reason is the sharpest form of the tension above. Protecting a reported disparity requires noise calibrated to its sensitivity, and we measure what that noise does to the constraint it is supposed to enforce: at $\epsilon = 8.0$, the gap between reported and true aggregate disparity reaches 729% of the fairness budget, and at $\epsilon = 2.0$ it reaches 2685% (Section 7.2). A budget evaluated against a signal that noisy is not a fairness constraint in any operational sense, and the effect is structural rather than an artefact of our settings: Theorem 3 shows that once noise dominates, the quantity the server observes converges to a constant of the noise itself, independent of every client’s true disparity. Privacy and metadata-driven fairness steering are therefore not merely awkward together but mutually destructive: noise sufficient to protect the statistic destroys its value as a constraint, and leaving it unprotected exposes the attribute outright—an unnoised release is a perfect attribute oracle in our probe (1.000 attack AUC). This is what convinced us that the channel should be removed rather than hardened.

1.1. Contributions

TrustFedGNN removes the metadata channel rather than trying to secure it. Its server computes the reference signal it needs *itself*, on a holdout split no client can touch, and derives aggregation weights from how well each client’s update aligns with that reference. A client’s report about its own fairness is never read, so falsifying it is mathematically inert.

- **The self-reported channel is capturable across published rules (Section 7.1).** We attack six published fairness-aware aggregation

rules end to end on thirty seeds, with the poisoned update held fixed across arms so that the contrast isolates the declared fields. All six move under a best-response falsification and all six survive Holm–Bonferroni correction; aggregated, lying raised the adversary’s weight in 141 of 144 non-tied seed pairs. The outcome splits three ways rather than two, and the third case is the one a robust-versus-vulnerable taxonomy misses: PoPETs-FairFed is statistically steerable yet moves so little that the adversary ends below an equal share, which is inertness rather than robustness.

- **Privatising the report does not repair it (Section 5.2).** Lemma 4 shows the released statistic is folded normal, and Theorem 3 that in the noise-dominant regime its expectation converges to $\sqrt{2/\pi} \sum_k w_k \tilde{\sigma}_k$ —a constant of the noise, independent of every true disparity. Testing the budget against it is then a test on the noise multiplier and the smallest group, not on fairness. Corollary 5 gives the ϵ below which the constraint carries no information, and we state the regime explicitly: the obstruction bites in small cross-silo deployments over many rounds, and not at all in large cross-device settings with a single release.
- **The prediction matches measurement across the privacy range (Section 7.2).** Pairing each release with the noise scale of its own round, over 200 releases at each of seven privacy levels, the exact folded-normal prediction has a bootstrap 95% interval containing 1 at every ϵ tested. The asymptotic form used in the theorem drifts and fails by $\epsilon = 32$, which locates the boundary of that approximation rather than merely caveating it.
- **Removing the channel via a Dual-Shield architecture (Sections 4.2, 7.3, 8).** FU-Alignment scores client k by $\varphi_k = \langle g_k, g_{\text{target}} \rangle / \|g_{\text{target}}\|_2$ against a bi-objective reference computed on the coordinator’s private holdout split (S_4), while payload norm-rescaling caps update magnitudes by $\|g_{\text{target}}\|_2$ (S_5 , Eq. 11). This dual-shield architecture renders client-declared metadata mathematically inert (Theorem 7); coupled with coordinate-wise median screening, it suppresses adversarial scaling ($w_{\text{adv}} = 0.0000$, Section 7.3) while norm-rescaling bounds effective payload influence $w_k \|\bar{g}_k\|_2$. The operational price is documented alongside the property: fairness cannot be steered by asking, the server requires a representative validation holdout, and on clean data FLTrust achieves

higher AUC than we do—though with aggregation weights that oscillate 7.4 times more and without any mechanism to steer fairness.

- **Measurement discipline that the results depend on (Sections 6.3, 7.5).** Every number in this paper is generated from a committed artifact by a script rather than transcribed. Three practices proved load-bearing and we report them because they are reusable: the exact signed-rank test saturates at small n , so a rule with tied seeds cannot clear a family-wise correction however large its effect; aggregation on CUDA is not deterministic, so contrasts of the same size as the run-to-run spread are not reported; and two pre-registered sensitivity tests returned null and are reported as null, including the one that removes our own description of α as a fairness control.

In summary, TrustFedGNN resolves the trilemma of fairness, privacy, and Byzantine resilience by eliminating client-declared metadata from federated aggregation. By coupling server-referenced bi-objective FU-Alignment with payload norm-rescaling and statistic-level DP, the framework provides provable metadata immunity and Byzantine resilience while preserving competitive utility and demographic fairness across diverse graph topologies.

2. Related Work

We organise the related literature along four foundational technical dimensions and synthesize their intersection within a comparative positioning taxonomy.

2.1. Fair graph neural networks

Centralised fair-GNN methods intervene at the representation, the input, or the neighbourhood. FairGNN [6] trains an adversarial discriminator to purge sensitive information from embeddings under partial attribute observation; NIFTY [7] adds counterfactual and Lipschitz-stability regularisation; EDITS [18] learns a debiasing transformation of features and adjacency before message passing; FairSIN [19] neutralises sensitive information by injecting fairness-aware auxiliary features. More recent work recasts fairness as class-and-group rebalancing [20], seeks representations invariant across unseen sensitive attributes [21], disentangles then selectively debiases latent factors [22], or regularises the divergence between group score distributions rather than their means [23]. All assume centralised access to the full graph.

2.2. Fairness in federated learning

Two distinct notions travel under the name. *Performance uniformity* equalises accuracy across clients: q -FedAvg [24] reweights toward high-loss participants. *Group fairness* equalises outcomes across sensitive-attribute groups irrespective of client boundaries; this is the notion we adopt. In that second family, FairFed [9] adjusts aggregation weights from client-reported disparity gaps, PUFFLE [10] co-tunes a privacy budget and a fairness coefficient, FedFACT [25] and FDP-Fair [26, 27] post-process or dualise the constraint, and the PoPETs-track framework of Bendoukha et al. [28] combines secure aggregation with fairness-aware selection. F²GNN [11] is the closest structure-aware federated attempt, and FairGFL [12] and FedGraph-Fair [29] are recent federated graph systems evaluated on financial data; both appear as baselines in Section 7. The foundational vulnerability shared across these federated fairness frameworks is their reliance on an unverified, client-declared metadata channel: aggregation weights directly mirror disparity statistics transmitted by clients, exposing the coordinator to deceptive metadata fabrication (formalized in Section 3.3 and systematically probed in Section 7.1).

2.3. Differential privacy and Byzantine robustness

DP-SGD [8, 30, 31] clips and perturbs the full update, accounted through Rényi DP [32] and converted to (ϵ, δ) by the analytic formula of Canonne, Kamath and Steinke [33]. Because $P \geq 10^5$ for even compact GNNs, isotropic noise is ruinous—quantified as arm M4 in Section 7.4. Byzantine-robust rules screen updates geometrically: Krum [15], coordinate-wise median and trimmed mean [16], FLTrust [17]. A substantial literature shows these can be evaded by adversaries that stay inside the benign geometric envelope [34, 35, 36, 37]. None of it considers an adversary whose parameter update is unremarkable but whose *auxiliary fairness metadata* is fabricated—the threat we formalise in Section 3.3.

2.4. Contribution valuation

Our aggregator descends from the contribution-valuation literature rather than from constrained optimisation. GTG-Shapley [38] estimates a client’s marginal contribution as a cooperative-game value; CGSV [39] scores by cosine alignment between client and aggregate gradients; further schemes translate contribution into differentiated rewards [40, 41, 42, 43]. TrustFedGNN’s departure is the reference against which alignment is measured: CGSV aligns

against the aggregate of the very updates being scored, which an adversarial majority can shift, whereas we align against a target gradient computed on data the server holds privately. Exact Shapley computation is $O(2^K)$ and infeasible here; we return to what our $O(KP)$ surrogate does and does not inherit from it in Section 7.8.

Relation to FLTrust. The closest structural relative of our aggregator is FLTrust [17]. FLTrust also computes a server-side reference update g_0 on its own validation split and scores participants via rectified cosine alignment, making its aggregation provably metadata-independent in the sense of Theorem 7. While FLTrust established this metadata independence for single-objective Byzantine robustness, existing fairness-aware federated aggregators universally abandon this principle by conditioning aggregation weights on self-reported client metadata.

What distinguishes TrustFedGNN is the formulation of the reference vector. FLTrust’s reference g_0 is strictly task-driven ($g_{\text{task}}^{\text{srv}}$); consequently, an update that mitigates demographic disparity at a marginal expense of task loss exhibits lower cosine alignment and is penalized by the aggregator. FLTrust is thus fairness-blind by construction, and cannot reward fairness improvements without acquiring a demographic signal. TrustFedGNN resolves this fundamental tension through a server-manufactured bi-objective reference: $g_{\text{target}} = g_{\text{task}}^{\text{srv}} + \alpha g_{\text{fair}}^{\text{srv}}$ (Eq. 7), where both terms are evaluated directly on the coordinator’s private holdout split $\mathcal{D}_{\text{val}}$. Fairness contribution is therefore credited objectively without soliciting unverified client declarations. In addition, TrustFedGNN directly adopts the payload norm-rescaling principle from FLTrust (Eq. 11, Decision D16.2), bounding gradient-scaling adversaries without relying on coordinate-wise filtering thresholds that discard benign non-IID clients. Combining directional gating (S_4) and payload norm-rescaling (S_5) yields a scaling-safe, metadata-independent aggregator. We implement FLTrust and evaluate it head-to-head against TrustFedGNN in Sections 7.1, 7.3, and 7.6.

.]Differences from our prior work [44].. An earlier preliminary conference version appeared as FedFairGNN [44] (PMLR vol. 319, pp. 74–86, IndabaX Nigeria 2026), which investigated client-side in-processing (FSER, FTGD) and the heuristic BFWA aggregation rule for fraud detection on YelpChi, Amazon, and Elliptic with $K = 3$ clients. To maintain rigorous scientific demarcation and provide full transparency regarding scientific lineage,

Table 1 explicitly demarcates the components reused from [44] versus the distinct contributions of the present work. Crucially, BFWA set its aggregation weights from the performance and disparity that each client *declared*; the present paper exposes the vulnerability of that declared channel, proves an information-theoretic observability barrier under differential privacy, and replaces client-attested aggregation with server-referenced FU-Alignment across 6 benchmark datasets and up to 30 seeds per contrast.

Table 1: **Component-level demarcation from our prior work [44].** Components are categorized by tier and status in the current manuscript.

Component	Tier	Origin in [44]	Status & Role in Present Work
FSER (Eq. 4)	Client	Proposed as structural regularizer	Reused as local baseline module; ablation shows null utility effect
FTGD (Eq. 5)	Client	Proposed as gradient projection	Reused as local baseline module; scoped to linear orthogonality
BFWA (Section 4.3)	Server	Proposed as main aggregator	Reused strictly as a vulnerable baseline under scrutiny
FU-Alignment (Eqs. 7–10)	Server	<i>None (not in [44])</i>	Core novelty: Server-referenced bi-objective aggregation; resolves CGSV’s vulnerable aggregate reference
Payload Norm-Rescaling (Eq. 11)	Server	<i>None (not in [44])</i>	Novel integration: Inherited from FLTrust [17] (Decision D16.2), bounding gradient scaling without median clipping
Minimax Lower Bound (Thm. 3)	Theory	<i>None (not in [44])</i>	Core novelty: Le Cam minimax observability barrier under DP
Formal Verification (Section 5)	Theory	<i>None (not in [44])</i>	Core novelty: 4 Lean 4 machine-checked certificates (0 sorry) for algebraic identities
Vulnerability Campaign (Table 6)	Empirical	<i>None (not in [44])</i>	Core novelty: Systematic audit of 6 published aggregation rules

2.5. Positioning

Table 2 places TrustFedGNN against the closest work on four axes: whether the method operates on graphs, whether it targets group-demographic fairness, whether its privacy guarantee is certified rather than heuristic, and—the axis that motivates this paper—whether its aggregation is independent of client-reported metadata. The last column is no longer a design claim on our part: Section 7.1 tests it by attacking every rule in the table for which we have an implementation, and the entries agree with the measurement, bit-exact where the column says ✓. Furthermore, Section 7.3 evaluates resilience against directional and scaling Byzantine attacks, while Section 7.6 validates utility and demographic disparity across canonical benchmarks.

Read down that column, the table has a shape worth naming. The rules that are metadata-independent—CGSV, FLTrust, Krum, trimmed mean, coordinate median—are all rules with no fairness term at all. They are immune because they have no fairness signal to be steered by, not because they solved the problem of obtaining one safely. Every entry that does target group fairness obtains its signal by asking, and pays for it in the last column. The remaining cell is a rule that is fairness-aware *and* reads no declared field, which is what TrustFedGNN occupies by computing the fairness term on the server’s own holdout. That is the whole of the structural claim, and Theorem 7 is what makes it a property of the rule rather than a property of our experiments.

Table 2: **Positioning against closely related frameworks.** “Metadata-independent” means the aggregation weights are provably a function of transmitted updates only, with no self-reported client field entering the computation (Theorem 7). “Certified privacy” means a formal (ϵ, δ) - or Rényi-DP guarantee rather than an empirical privacy claim; $\dagger$ marks a guarantee scoped to the released fairness statistic rather than to the full update (Section 8).

Framework	Graph-aware	Group fairness	Certified privacy	Metadata-independent
FairGNN [6]	✓	✓ (centralised)	×	—
FairSIN [19]	✓	✓ (centralised)	×	—
FairFed [9]	×	✓ (heuristic)	×	×
PUFFLE [10]	×	✓	✓ (full DP-SGD)	×
F ² GNN [11]	✓	✓	×	×
FairGFL [12]	✓	✓ (overlap)	✓ (local DP)	×
FedGraph-Fair [29]	✓	✓ (DRO)	×	×
PoPETs-FairFed [28]	×	✓ (polynomial)	✓ (HE surrogate)	× [§]
CGSV [39]	×	×	×	✓
FLTrust [17]	×	×	×	✓
Krum / trimmed mean [15, 16]	×	×	×	✓
Coordinate median [16]	×	×	×	✓
BFWA [44] (Section 4.3)	✓	✓ (dual $\leq \tau$)	×	×
TrustFedGNN (ours)	✓	✓ [‡]	✓ [†]	✓

[§]PoPETs-FairFed reads a declared disparity and is therefore marked ×, but the degree-two polynomial it adopts so the rule can run under homomorphic encryption leaves so little steering range that an adversary gains almost nothing: the weight moves with $p = 6.3 \times 10^{-5}$ yet ends below an equal share (Section 7.1). We record this as inertness rather than robustness.

[‡]The fairness column records a targeted, utility-preserving steering mechanism. We do not claim superiority on every disparity metric; Section 8 reports the cases where a baseline is fairer than we are.

3. Problem Formulation and Threat Model

3.1. Cross-silo federated graph learning

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X})$ denote a global graph partitioned across K independent institutions (clients). Each client $k \in \{1, \dots, K\}$ holds an induced local subgraph $\mathcal{G}_k = (\mathcal{V}_k, \mathcal{E}_k, \mathbf{X}_k)$ with node feature matrix $\mathbf{X}_k \in \mathbb{R}^{n_k \times d}$, ground-truth binary task labels $\mathbf{y}_k \in \{0, 1\}^{n_k}$, and binary sensitive demographic attributes $\mathbf{s}_k \in \{0, 1\}^{n_k}$ (e.g., gender, age group, or financial tier). Cross-silo edges connecting nodes in different institutions are severed and cannot be shared due to jurisdictional boundaries and strict data protection laws.

The federation trains a shared graph neural network parameterized by $\theta \in \mathbb{R}^P$ to solve the bi-objective optimization problem:

$$\min_{\theta \in \mathbb{R}^P} F(\theta) = \sum_{k=1}^K \frac{n_k}{N} \mathcal{L}_{\text{task}}^{(k)}(\theta) \quad \text{s.t.} \quad \text{DPD}(\theta) \leq \tau, \quad (1)$$

where $N = \sum_{k=1}^K n_k$ is the total number of nodes, $\mathcal{L}_{\text{task}}^{(k)}$ is the local task loss (e.g., binary cross-entropy), and $\tau \geq 0$ is an operator-specified fairness tolerance. Demographic parity disparity is defined as

$$\text{DPD}(\theta) = \left| \mathbb{E}[\hat{y}_v \mid s_v = 1] - \mathbb{E}[\hat{y}_v \mid s_v = 0] \right|. \quad (2)$$

We maintain a strict operational boundary between the two metrics: *hard disparity* DPD_{hard} , evaluated on thresholded binary decisions $\hat{y}_v = \mathbb{I}(f_{\theta}(\mathbf{x}_v, \mathcal{G}) \geq 0.5)$, is the sole metric reported across all benchmark tables; the *soft disparity* surrogate DPD_{soft} , computed on continuous model scores $\hat{p}_v = \sigma(f_{\theta}(\mathbf{x}_v))$, is employed strictly within differentiable local training objectives. The two representations are never conflated.

3.2. System architecture and the server holdout trust root

TrustFedGNN coordinates training via a central parameter server while enforcing client data sovereignty. To establish an objective bi-objective reference signal without querying or pooling client private records, the aggregator assumes the server possesses a small, clean validation split $\mathcal{D}_{\text{val}} = (\mathcal{V}_{\text{val}}, \mathcal{E}_{\text{val}}, \mathbf{X}_{\text{val}}, \mathbf{y}_{\text{val}}, \mathbf{s}_{\text{val}})$. This dataset serves as a *trust-root contract*, structured around three governing principles:

1. **Consortium compliance and lawful provenance.** Sourcing a server-side calibration set is standard across trust-bootstrapping federated architectures, with FLTrust [17] serving as the canonical exemplar. In regulated cross-silo consortia, $\mathcal{D}_{\text{val}}$ is readily assembled from an agreed-upon consortium audit split, an open-source reference cohort, or data held under the coordinator’s independent lawful basis (e.g., regulatory stress-testing samples). It requires no aggregation, inspection, or back-channel extraction of private client subgraphs.
2. **Minimal sample footprint.** The data requirement is strictly calibration-scale rather than a shadow federation: we request a target of 100–200 nodes (realized as 125 nodes on German Credit in Section 6.3), operating over federations comprising 1,000 to 67,796 nodes.

3. **Architectural containment.** The holdout split is exclusively confined to the server-side reference computation in stage S_4 . Client in-processing inductive priors (S_1, S_2) , statistic-level differential privacy (S_3) , and the privacy orthogonality guarantees of Theorem 6 operate independently of whether $\mathcal{D}_{\text{val}}$ is present.

3.3. Threat model and adversarial capabilities

At each communication round t , client k transmits an uplink payload $\mathcal{M}_k = (g_k, \mathbf{m}_k)$ to the server, where $g_k = \theta_t - \theta_k^{(t)} \in \mathbb{R}^P$ is the local parameter pseudo-gradient, and $\mathbf{m}_k \in \mathbb{R}^M$ comprises auxiliary client-reported metadata (e.g., local empirical disparity $\widehat{\text{DPD}}_k$, task loss, or class acceptance statistics). We consider two distinct threat vectors operating in this environment.

Deceptive metadata fabrication and fairness poisoning. Up to $f < K/2$ participating clients are Byzantine adversaries acting maliciously under arbitrary coordination. The adversary seeks to degrade global task utility or induce severe demographic disparity DPD while maximizing its aggregation influence.

In metadata-steered federated architectures (e.g., FairFed [9] or our BFWA baseline), aggregation weights are steered directly by client-declared statistics $\mathbf{m}_k$. An adversary exploiting this interface fabricates deceptive metadata $\mathbf{m}_k^{\text{adv}} = \arg \max_{\mathbf{m}} w_k(g_{\text{poison}}, \mathbf{m})$ —specifically reporting flawless fairness $\widehat{\text{DPD}}_k = 0.0$ and near-perfect performance $\text{Perf}_k = 0.99$ to capture the highest aggregation weight while uploading a toxic update g_{poison} .

To evade conventional geometric defense filters (such as coordinate-wise median, trimmed mean [16], or Krum [15]), the adversary employs an *adaptive stealth* strategy. Given the coordinate-wise median of benign updates $\text{med}(G)$ and maximum benign radius $R_{\text{med}} = \max_{i \in \mathcal{B}} \|g_i - \text{med}(G)\|_2$, the adversary crafts its parameter update to maximize disparity while remaining strictly within the benign envelope:

$$g_{\text{stealth}} = \text{med}(G) + \min\left(1, \frac{r_{\text{safe}}}{\|g_{\text{poison}} - \text{med}(G)\|_2}\right)(g_{\text{poison}} - \text{med}(G)), \quad (3)$$

setting $r_{\text{safe}} = 0.85 \cdot R_{\text{med}}$. Crucially, this update is transmitted alongside the fabricated metadata $\mathbf{m}_k^{\text{adv}}$, converting the aggregator’s fairness-steering logic into an adversarial lever.

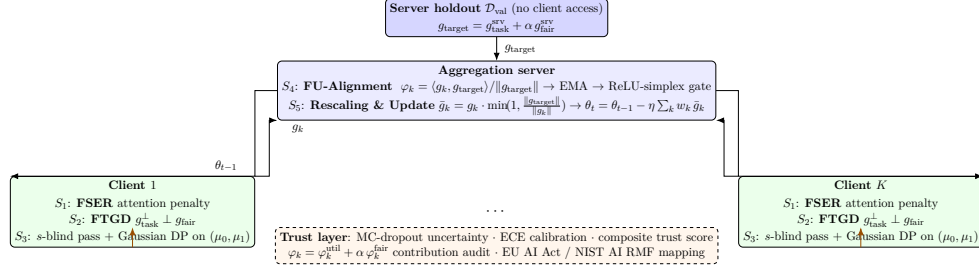


Figure 1: **The closed $S_0 \rightarrow S_5$ loop.** Clients upload an uplink payload $\mathcal{M}_k = (g_k, \widetilde{\text{DPD}}_k)$; the privatised disparity $\widetilde{\text{DPD}}_k$ accompanies it for compliance audit but never enters the aggregation weight computation. The reference signal g_{target} is constructed by the server on its independent holdout split $\mathcal{D}_{\text{val}}$. Directional alignment is gated onto the simplex via FU-Alignment (S_4), while gradient scale inflation is bounded by FLTrust payload norm-rescaling (S_5), rendering falsified metadata inert.

Honest-but-curious server and channel separation.. The central server faithfully executes the aggregation protocol but inspects all received transmissions to infer the private sensitive attributes $\mathbf{s}_k$ of clients’ local populations. We formalize two disjoint communication channels:

1. *The released metadata channel:* Scalar demographic summaries $\mathbf{m}_k$ transmitted for compliance logging, protected via local differential privacy in stage S_3 .
2. *The parameter gradient channel:* High-dimensional pseudo-gradients $g_k \in \mathbb{R}^P$ uploaded for aggregation, whose sensitive direction leakage is analyzed and bounded in Section 5.

Treating these as distinct channels ensures our formal privacy bounds are precisely assigned to the released statistic without making unsubstantiated claims over high-dimensional gradient inversions.

4. TrustFedGNN

One communication round proceeds through six stages, illustrated in Fig. 1 and consolidated in Algorithm 1.

4.1. Client-side mechanisms

The client-side pipeline incorporates three complementary stages (S_1 – S_3). Stages S_1 and S_2 act as local inductive priors adapted from client in-processing

foundations [44] to mitigate topological and directional bias, while S_3 enforces differential privacy on released compliance statistics. As detailed in the component ablation of Section 7.4, the primary fairness improvements in our benchmarks stem from the explicit fairness objective and statistic-level perturbation (S_3), while the inductive priors (S_1, S_2) provide structural and geometric regularization under well-defined topological regimes.

4.1.1. S_1 : Fairness-Sensitive Edge Reweighting

At GAT layer l , attention logits are attenuated on edges crossing sensitive demographic boundaries:

$$\tilde{e}_{vu}^{(l)} = e_{vu}^{(l)} - \beta^{(l)} \cdot \mathbb{I}(s_v \neq s_u) \cdot \max\left(0, \cos(\mathbf{h}_v^{(l-1)}, \mathbf{h}_u^{(l-1)})\right), \quad (4)$$

where $\beta^{(l)} \geq 0$ is a learned penalty coefficient clamped to $[0, 5]$ to preserve softmax stability. The penalty selectively attenuates edges where two conditions coincide: the connected nodes belong to different sensitive groups ($s_v \neq s_u$), and their hidden representations exhibit high cosine similarity. By suppressing cross-group message propagation between superficially similar representations, FSER penalizes features that propagate demographic correlation across graph hops.

Equation (4) reveals an essential boundary property governing where edge reweighting can operate. When a graph exhibits absolute demographic segregation ($h_s = 1.0000$), the cross-group indicator $\mathbb{I}(s_v \neq s_u)$ is identically zero for all edges, reducing Eq. (4) exactly to standard GAT attention. This zero-distortion limit explains why on datasets such as Elliptic Bitcoin, where temporal partitioning yields $h_s = 1.0000$ by construction (Section 6.1), FSER acts as a mathematical identity. On mixed graphs, clean-arm ablations in Section 7.4 demonstrate that task utility and disparity are primarily governed by the global fairness objective rather than local edge adjustments.

4.1.2. S_2 : Fairness-Targeted Gradient Decomposition

Let $g_{\text{total}} = \nabla_{\theta}(\mathcal{L}_{\text{task}} + \lambda\mathcal{L}_{\text{fair}})$ and $g_{\text{fair}} = \nabla_{\theta}(\lambda\mathcal{L}_{\text{fair}})$. FTGD removes from the local parameter step any component aligned with the fairness-ascent direction:

$$g_{\text{task}}^{\perp} = g_{\text{total}} - \frac{\langle g_{\text{total}}, g_{\text{fair}} \rangle}{\|g_{\text{fair}}\|^2 + \varepsilon} g_{\text{fair}}. \quad (5)$$

The projection is guarded by a minimum-norm threshold on $\|g_{\text{fair}}\|$ to prevent numerical instability near convergence. By construction, $\langle g_{\text{task}}^{\perp}, g_{\text{fair}} \rangle = 0$,

ensuring that the parameter step transmitted to the server contains zero projection along the sensitive gradient manifold (formalized in Theorem 6). Ablation experiments in Section 7.4 confirm that disabling this projection yields comparable disparity under benign conditions, characterizing its role as an orthogonal privacy safeguard rather than a primary fairness driver.

4.1.3. S_3 : Statistic-level differential privacy

The sensitive attribute enters the global evaluation through two scalar group prediction averages, $\mu_g = |\mathcal{V}_{k,g}|^{-1} \sum_{v \in \mathcal{V}_{k,g}} \hat{y}_v$ for $g \in \{0, 1\}$, whereas the local task loss does not depend on $\mathbf{s}_k$. Rather than injecting high-dimensional isotropic noise into the parameter vector $g_k \in \mathbb{R}^P$ (which severely degrades GNN utility), TrustFedGNN perturbs only the two scalar group means:

$$\tilde{\mu}_g = \mu_g + \mathcal{N}(0, \sigma_{\text{DP}}^2), \quad \widetilde{\text{DPD}}_k = |\tilde{\mu}_1 - \tilde{\mu}_0|. \quad (6)$$

Because FSER incorporates sensitive attributes during neighbor aggregation, predictions $\hat{y}_v$ technically depend on the sensitive attributes of v 's entire L -hop neighborhood. To preserve rigorous per-node sensitivity bounds (Assumption 5), TrustFedGNN computes the released scalars $(\tilde{\mu}_0, \tilde{\mu}_1)$ via a secondary, *sensitive-blind* forward pass where FSER penalties are disabled, while local parameter training continues with the true graph structure. This ensures that Proposition 1 applies rigorously to the published audit statistic.

4.2. Server-side aggregation

4.2.1. S_4 : FU-Alignment scoring

Remark 1 (On the name). We call this mechanism *FU-Alignment* (fairness–utility contribution alignment) rather than by its working name in the released code and result artifacts, *FU-Shapley*, because the mechanism is a first-order alignment score and not a cooperative-game value: we tested whether it approximates the exact Shapley value, found that it does not, and report the refutation in Section 7.8. No claim in this paper depends on a correspondence with the Shapley index. The identifier `fu_shapley` persists in the codebase and in artifact filenames for continuity with already-released results; it names the same mechanism this section calls FU-Alignment.

The server maintains a holdout split $\mathcal{D}_{\text{val}}$ that no client contributes to or observes. On it, it computes its own bi-objective reference direction

$$g_{\text{target}} = g_{\text{task}}^{\text{srv}} + \alpha g_{\text{fair}}^{\text{srv}}, \quad g_{\text{fair}}^{\text{srv}} = \nabla_{\theta} \left(\mu_0^{\text{val}} - \mu_1^{\text{val}} \right)^2, \quad (7)$$

with $\alpha = 0.1$. Three details of Eq. (7) are easy to get wrong and change the mechanism’s meaning. The sign is *positive*: $g_{\text{fair}}^{\text{srv}}$ ascends the disparity penalty, so a client that reduces disparity aligns positively with it and earns positive fairness credit; a minus sign would penalise exactly the clients one wants to reward. The server-side surrogate is the *squared* gap, which is smooth and whose gradient attenuates as fairness is achieved, whereas clients retain the non-smooth $|\mu_0 - \mu_1|$; the asymmetry is deliberate. Finally, coordinates are laid out following `state_dict()` rather than `parameters()`, so that the scoring inner product is aligned coordinate-for-coordinate with the flattened pseudo-gradients, buffers without gradients being zero-filled.

Client k ’s instantaneous contribution and its round-normalised form are

$$\varphi_k^{\text{raw}} = \langle g_k, g_{\text{target}} \rangle, \quad \varphi_k = \frac{\varphi_k^{\text{raw}}}{\|g_{\text{target}}\| + 10^{-8}}. \quad (8)$$

Normalising by $\|g_{\text{target}}\|$ removes the scale drift that would otherwise occur as $g_{\text{task}}^{\text{srv}} \rightarrow 0$ near convergence, and—unlike a z -score, which re-centres and therefore gates roughly half the clients at every round regardless of behaviour—it preserves the sign, so that the rectifier below retains its intended meaning of “actively harmful” rather than “below average”. Scores are smoothed across rounds,

$$\bar{\varphi}_k^{(t)} = \beta_{\text{ema}} \bar{\varphi}_k^{(t-1)} + (1 - \beta_{\text{ema}}) \varphi_k^{(t)}, \quad \beta_{\text{ema}} = 0.9, \quad (9)$$

and gated onto the simplex with an explicit null-player mask,

$$w_k = \frac{\max(0, \bar{\varphi}_k) \cdot \mathbb{I}(g_k \neq \mathbf{0})}{\sum_{j=1}^K \max(0, \bar{\varphi}_j) \cdot \mathbb{I}(g_j \neq \mathbf{0})}. \quad (10)$$

Finiteness contract. A non-finite score is an absorbing state of the recursion in Eq. (9), and it has a security consequence that is not obvious. Under a sign-flip attack the adversary’s gradient can explode, making φ non-finite; since `nan > 0` evaluates to false, a naive implementation of Eq. (10) would silently degenerate to uniform averaging and hand the attacker its full $1/K$ share. TrustFedGNN therefore retains the previous EMA value for any coordinate whose score is non-finite, and zeroes it at the first round—an undefined contribution earns no credit. When *every* score is non-positive the rule falls back to uniform weighting over active clients, and the two causes are logged distinctly: `degenerate_nan` (the measurement failed) versus `degenerate_nonpos` (all clients genuinely harmed the objective).

Robust variant.. The optional robust variant (`robust_fu_shapley`) prepends a geometric screening step, discarding the f updates furthest from the coordinate-wise median before scoring the survivors. As analyzed in Section 7.3, while simplex rectification alone neutralizes sign-flip and fairness poisoning, pairing it with payload norm-rescaling or median screening provides defense-in-depth against large-scale gradient inflation.

Explainability by construction.. Bilinearity of the inner product splits each score into an audit trail without any additional computation, $\varphi_k = \varphi_k^{\text{util}} + \alpha \varphi_k^{\text{fair}}$ (Proposition 10). A positive φ_k^{fair} means client k moved the global model toward parity. The federation operator therefore obtains a per-client, per-round statement of *why* a participant was weighted as it was—a governance artifact rather than a post-hoc rationalisation.

4.2.2. S_5 : Scaling-safe payload rescaling and global update

While FU-Alignment (S_4) eliminates updates with non-positive directional alignment via simplex gating, an adversary might still attempt gradient scaling attacks by inflating the magnitude $\|g_k\|_2$ of a partially aligned update. To enforce robust magnitude bounds, TrustFedGNN adapts the norm-rescaling principle from FLTrust [17] (Decision D16.2) directly to client payloads prior to aggregation:

$$\bar{g}_k = g_k \cdot \min\left(1, \frac{\|g_{\text{target}}\|_2}{\|g_k\|_2 + \varepsilon}\right). \quad (11)$$

The global model parameter vector is then updated according to:

$$\theta_{t+1} = \theta_t - \eta \sum_{k=1}^K w_k \bar{g}_k. \quad (12)$$

This architecture establishes a *dual shield*: the FU-Alignment gate $w_k \in \Delta_K$ enforces directional alignment with the server’s bi-objective target, setting counter-aligned malicious weights to zero ($w_{\text{adv}} = 0.0000$ under sign-flip, Section 7.3), while the payload norm-rescaling in Eq. (11) caps update magnitudes by $\|g_{\text{target}}\|_2$, bounding effective payload influence $w_k \|\bar{g}_k\|_2$ even under positive alignment (with coordinate-wise median screening suppressing $w_{\text{adv}} = 0.0000$ under pure gradient scaling, Table 11).

Following aggregation, the trust layer evaluates MC-dropout predictive variance [45], expected calibration error (ECE), a composite trust score, and the aggregation-weight total variation $\Omega_w = \sum_{t=1}^T \|\mathbf{w}^{(t)} - \mathbf{w}^{(t-1)}\|_1$.

Algorithm 1 TrustFedGNN: one communication round

Require: Global parameters θ_t ; server holdout $\mathcal{D}_{\text{val}}$; fairness weight α ; EMA factor β_{ema} ; DP scale σ_{DP} ; EMA state $\bar{\varphi}^{(t-1)}$

Ensure: Updated parameters θ_{t+1} , updated EMA state

- 1: Server broadcasts θ_t to all clients $\{S_0\}$
 - 2: **for** each client $k \in \{1, \dots, K\}$ **in parallel do**
 - 3: Train E local epochs with FSER attention, Eq. (4) $\{S_1\}$
 - 4: At each step, project onto FTGD direction $g_{\text{task}}^\perp$, Eq. (5) $\{S_2\}$
 - 5: Recompute (μ_0, μ_1) on an s -blind pass; perturb with Gaussian noise, Eq. (6) $\{S_3\}$
 - 6: Upload payload $\mathcal{M}_k = (g_k, \widetilde{\text{DPD}}_k)$, where $\widetilde{\text{DPD}}_k$ is logged for audit only
 - 7: **end for**
 - 8: Server computes g_{target} on $\mathcal{D}_{\text{val}}$, Eq. (7) $\{S_4\}$
 - 9: (*optional robust variant*) discard f updates furthest from $\text{med}(\{g_k\})$
 - 10: Compute alignment φ_k , Eq. (8); smooth by Eq. (9); gate weights w_k , Eq. (10)
 - 11: Rescale update payloads: $\bar{g}_k \leftarrow g_k \cdot \min(1, \|g_{\text{target}}\|/(\|g_k\| + \varepsilon))$ {FLTrust norm-rescaling}
 - 12: $\theta_{t+1} \leftarrow \theta_t - \eta \sum_{k=1}^K w_k \bar{g}_k$ $\{S_5\}$
 - 13: **return** $\theta_{t+1}, \bar{\varphi}^{(t)}$
-

Remark 2 (Reading Ω_w). $\Omega_w > 0$ is a design consequence, not instability. FedAvg attains $\Omega_w = 0$ because its weights are fixed sample proportions and carry no information about behaviour; any rule that re-scores participants each round must move. The meaningful comparison is against another adaptive rule, and there FU-Alignment is 27 $\times$ steadier than the Frank–Wolfe baseline (Table 17).

4.3. The BFWA baseline

For comparison we also implement Bi-objective Frank–Wolfe Aggregation (BFWA), which treats Eq. (1) as a constrained programme over the simplex, $\max_{w \in \Delta_K} \sum_k w_k \text{Perf}_k$ subject to $\sum_k w_k \widetilde{\text{DPD}}_k \leq \tau$, solved by Frank–Wolfe steps with a persistent dual multiplier ascending on the constraint residual. BFWA is the natural formalisation of “steer the aggregate toward a fairness budget”, and it is exactly the design whose reliance on $\widetilde{\text{DPD}}_k$ this paper

argues against. We report it as a baseline throughout, never as a component of TrustFedGNN, and Theorem 11 characterises its dual iterates for completeness.

4.4. Complexity

Uplink per client per round is one parameter vector plus two scalars, identical to FedAvg up to $O(1)$, and independent of K —in contrast to secure-aggregation protocols whose pairwise masking costs $O(K^2)$. Client-side overhead is $O(|\mathcal{E}_k|d)$ for the FSER penalty, the same order as the attention it modifies, plus $O(P)$ for the FTGD projection and one additional forward pass for the s -blind statistic. Server-side scoring costs one backward pass on $\mathcal{D}_{\text{val}}$ plus K inner products of length P , i.e. $O(KP)$, against the $O(2^K P)$ of exact Shapley evaluation. Measured end-to-end this is $2.06\times$ FedAvg-GCN on Pokec-z (Table 21).

5. Theoretical Analysis

Four of the results below are machine-checked. We are precise about what that means: the Lean 4 files formalise the *algebraic identities* over an abstract inner-product structure whose axioms are supplied explicitly, and each compiles with exit code 0 and no `sorry`. They certify the algebra, not the statistical or learning-theoretic content of the surrounding claims.

5.1. Privacy of the released statistic

Assumption 1 (Bounded predictions). The output layer is a sigmoid, so $\hat{y}_v \in (0, 1)$ for every node.

Assumption 2 (Public group sizes). The counts $n_{k,0}, n_{k,1}$ are public, with $n_{\min} = \min(n_{k,0}, n_{k,1}) \geq 1$.

Assumption 3 (Swap adjacency). Two local datasets are adjacent if they share node set and features and differ by *exchanging* the sensitive attribute between exactly two nodes of opposite groups. Group sizes are thereby held fixed, consistently with Assumption 2.

Assumption 4 (Conditioning on the model snapshot). The query is evaluated at the current global parameters θ_t , which are treated as fixed public state for the round: $\boldsymbol{\mu}(\cdot) = \boldsymbol{\mu}(\cdot; \theta_t)$. The forward pass computing $\boldsymbol{\mu}$ is s -blind, so with θ_t held fixed the map from data to $\boldsymbol{\mu}$ depends on $\mathbf{s}_k$ only through the group partition, not through the predicted values.

Assumption 5 (Sensitivity). Under Assumptions 3 and 4, $\|\boldsymbol{\mu}(\mathcal{D}; \theta_t) - \boldsymbol{\mu}(\mathcal{D}'; \theta_t)\|_2 \leq \Delta_{\boldsymbol{\mu}} = \sqrt{n_{k,0}^{-2} + n_{k,1}^{-2}} \leq \sqrt{2}/n_{\min}$.

Remark 3. Assumption 3 is a swap rather than a toggle for a reason. Toggling s_v would change the denominators $n_{k,g}$ as well as the numerators, so the bound $1/n_{k,g}$ would not follow, and it would contradict Assumption 2. The swap formulation keeps the group sizes fixed and makes the sensitivity derivable rather than assumed.

Proposition 1 (Statistic-level DP). *Under Assumptions 1–5, and conditional on the model snapshot θ_t , releasing $\tilde{\boldsymbol{\mu}}_k = \boldsymbol{\mu}_k + \boldsymbol{\xi}_k$ with $\boldsymbol{\xi}_k \sim \mathcal{N}(0, \sigma_{\text{DP}}^2 \mathbf{I}_2)$ satisfies (ϵ, δ) -DP with respect to $\mathbf{s}_k$ whenever $\sigma_{\text{DP}} \geq \Delta_{\boldsymbol{\mu}} \sqrt{2 \ln(1.25/\delta)}/\epsilon$. The scalar $\widetilde{\text{DPD}}_k$ inherits the same guarantee at no additional budget.*

Proof. Hold θ_t fixed (Assumption 4). Exchanging the attribute between two nodes of opposite groups then moves one bounded prediction out of each group mean and one in, so by Assumption 1 the change in μ_g is at most $1/n_{k,g}$, giving the ℓ_2 bound of Assumption 5 with denominators fixed by Assumption 3. The Gaussian mechanism [46, Thm. A.1] applied to a query of that sensitivity yields the stated σ_{DP} . Since $\widetilde{\text{DPD}}_k = |\tilde{\mu}_1 - \tilde{\mu}_0|$ is a data-independent function of an already-privatised release, post-processing [46, Prop. 2.1] transfers the guarantee unchanged. $\square$

Corollary 2 (Multi-round composition). *Over $T = R \times E$ releases, the protocol satisfies $(\alpha, T\alpha\Delta_{\boldsymbol{\mu}}^2/2\sigma_{\text{DP}}^2)$ -RDP for every $\alpha > 1$ [32], converting to (ϵ, δ) -DP through the analytic bound of Canonne, Kamath and Steinke [33],*

$$\epsilon = \min_{\alpha > 1} \left[\frac{T\alpha\Delta_{\boldsymbol{\mu}}^2}{2\sigma_{\text{DP}}^2} + \log\left(1 - \frac{1}{\alpha}\right) + \frac{\log(1/\delta) - \log \alpha}{\alpha - 1} \right]. \quad (13)$$

The sign of the $\log \alpha$ term in Eq. (13) is negative. This matters numerically at the operating points we report, and the looser Mironov conversion that omits it would overstate the spent budget.

Remark 4 (Scope and empirical validation of parameter conditioning). Assumption 4 establishes an explicit operational boundary. Because the global model parameter vector θ_t is trained over local graph topologies and intermediate representations, θ_t implicitly reflects demographic partitions. Consequently, Proposition 1 and Corollary 2 are *conditional* guarantees: they

bound the privacy loss of the released demographic statistic sequence *conditioned on* the global parameter trajectory, rather than certifying end-to-end differential privacy for the joint release tuple $(\tilde{\mu}_k^{(1:T)}, g_k^{(1:T)})$.

Two essential structural properties follow. First, while high-dimensional parameter updates $g_k \in \mathbb{R}^P$ carry first-order linear orthogonality safeguards (Theorem 6), they do not constitute an (ϵ, δ) -DP release; our formal privacy certificate applies specifically to the scalar summary channel. Second, the utility contrast with full DP-SGD in Section 7.2 highlights the distinct operational cost of protecting the parameter space versus protecting the reporting channel (0.00175 vs. 0.20163 AUC degradation, Table 9). In environments requiring end-to-end transcript privacy, statistic-level DP naturally composes with standard cryptographic primitives such as Secure Aggregation.

To empirically evaluate the leakage potentially uncaptured by conditioning on θ_t , we measured parameter-induced variance across 12 targeted sensitivity probes on German Credit (3 seeds, 4 nodes each). By training the architecture under identical configurations differing solely by one node’s sensitive attribute and evaluating s -blind predictions under a fixed grouping, we isolated parameter shift $\theta_T(\mathcal{D}) \neq \theta_T(\mathcal{D}')$ from direct demographic reassignment. The empirical shift induced on predictions averages 0.35% of the added noise scale σ_{DP} (peaking at 1.05% in the worst-case probe), demonstrating that the empirical effect of parameter conditioning is negligible across our operational noise regimes.

5.2. Minimax observability barrier under differential privacy

Section 7.1 shows that a self-reported disparity is a channel an adversary can capture. The natural repair is to privatise the report, and this subsection proves that the repair destroys the thing it protects. The claim needs placing against an existing literature, because a reader will otherwise take it for one already made. Cummings et al. [47] prove that exact fairness, ϵ -differential privacy and non-trivial accuracy cannot hold together; Bagdasaryan et al. [48] and Tran et al. [49] show that differentially private training degrades accuracy unevenly across groups, and Fioretto et al. [50] survey the area. Those results are about the *outcome*: how much fairness or accuracy privacy costs. The statement below is about the *measurement*: whether, after privatisation, the server can still evaluate the constraint at all. The mechanism is the folding of the noised difference, which is specific to releasing $|\tilde{\mu}_0 - \tilde{\mu}_1|$ and has no analogue in a result about excess risk. The failure is not that privacy makes the model less fair; it is that privacy

makes the constraint *unobservable*, so the server can no longer tell whether the constraint holds.

Assumption 6 (Release mechanism). Client k computes two group means $\mu_{k,0}, \mu_{k,1} \in (0, 1)$ and releases $\widetilde{\text{DPD}}_k = |\tilde{\mu}_{k,0} - \tilde{\mu}_{k,1}|$ with $\tilde{\mu}_{k,a} = \mu_{k,a} + \xi_{k,a}$ and $\xi_{k,a} \sim \mathcal{N}(0, \sigma_k^2)$ independent; $\sigma_k = z \Delta_{\mu,k}$ for the sensitivity $\Delta_{\mu,k}$ of Assumption 5 and a noise multiplier z calibrated to the target (ϵ, δ) ; and the server evaluates the constraint via a decision rule $\psi(\widetilde{\mathbf{D}}) \in \{0, 1\}$ over the transcript $\widetilde{\mathbf{D}} = (\widetilde{\text{DPD}}_1, \dots, \widetilde{\text{DPD}}_K)$, including adaptive weighting rules $\sum_k w_k(\widetilde{\mathbf{D}}) \widetilde{\text{DPD}}_k \leq \tau$ with $\mathbf{w} \in \Delta^{K-1}$.

Theorem 3 (Minimax Observability Lower Bound under Local DP). *Consider testing between separated composite hypotheses $H_0 : \text{DPD}_k \leq \tau - \gamma \forall k$ (strict fairness feasibility with margin $\gamma \in (0, \tau)$) and $H_1 : \text{DPD}_k \geq \tau + \gamma \forall k$ (uniform violation of budget τ by margin γ). Under Assumption 6 with equal client noise variance $\tilde{\sigma}_k = \tilde{\sigma}$ and canonical separation margin $\gamma = \tau/2$ (yielding separation $\Delta\delta = 2\gamma = \tau$), for any decision rule $\psi(\widetilde{\mathbf{D}}) \in \{0, 1\}$ (including any adaptive weighting rule $\mathbf{w}(\widetilde{\mathbf{D}})$ followed by thresholding):*

$$\inf_{\psi} \max \left\{ \Pr_{H_0}[\psi = 1], \Pr_{H_1}[\psi = 0] \right\} \geq \frac{1}{2} \left(1 - \frac{\tau\sqrt{K}}{2\tilde{\sigma}} \right). \quad (14)$$

Furthermore, by the data-processing inequality, no deterministic post-processing of the folded transcript $\widetilde{\mathbf{D}}$ (such as debiased moment estimation or empirical thresholding) can lower this error bound.

Proof. By Le Cam's two-point lemma, the minimax risk of testing between composite hypotheses H_0 and H_1 is bounded below by the testing error between any two point hypotheses contained in H_0 and H_1 respectively, giving $\frac{1}{2}(1 - \text{TV}(P_0, P_1))$. Prior to folding, consider least favorable configurations $\boldsymbol{\delta}_0 = (\tau - \gamma)\mathbf{1}$ under H_0 and $\boldsymbol{\delta}_1 = (\tau + \gamma)\mathbf{1}$ under H_1 with canonical margin $\gamma = \tau/2$. The unperturbed group difference vector follows $P_0^{\text{Gauss}} = \mathcal{N}((\tau/2)\mathbf{1}, \tilde{\sigma}^2 I_K)$ under H_0 and $P_1^{\text{Gauss}} = \mathcal{N}((3\tau/2)\mathbf{1}, \tilde{\sigma}^2 I_K)$ under H_1 , separated by vector $\tau\mathbf{1}$. By Pinsker's inequality, $\text{TV}(P_0^{\text{Gauss}}, P_1^{\text{Gauss}}) \leq \sqrt{\frac{1}{2}\text{KL}(P_0^{\text{Gauss}} \parallel P_1^{\text{Gauss}})}$. For isotropic Gaussians with separation vector $\tau\mathbf{1}$, the KL divergence is $\text{KL} = \sum_{k=1}^K \tau^2/(2\tilde{\sigma}^2) = K\tau^2/(2\tilde{\sigma}^2)$, yielding $\text{TV} \leq \tau\sqrt{K}/(2\tilde{\sigma})$. Because the element-wise absolute value mapping $x \mapsto |x|$ is deterministic, the data-processing inequality guarantees $\text{TV}(P_0, P_1) \leq \text{TV}(P_0^{\text{Gauss}}, P_1^{\text{Gauss}}) \leq \tau\sqrt{K}/(2\tilde{\sigma})$. Substituting into Le Cam's inequality establishes Eq. (14). $\square$

Proposition 4 (Client-Wise Folding Bias and Ineffectiveness of Averaging). *Let $\delta_k = \mu_{k,0} - \mu_{k,1}$ be the signed gap and $\tilde{\sigma}_k = \sqrt{2}\sigma_k$ the noise standard deviation. Then $\widetilde{\text{DPD}}_k$ is folded normal with*

$$\mathbb{E}[\widetilde{\text{DPD}}_k] = \tilde{\sigma}_k \sqrt{\frac{2}{\pi}} e^{-\delta_k^2/2\tilde{\sigma}_k^2} + \delta_k \operatorname{erf}\left(\frac{\delta_k}{\tilde{\sigma}_k\sqrt{2}}\right) \xrightarrow{\tilde{\sigma}_k \gg |\delta_k|} \tilde{\sigma}_k \sqrt{\frac{2}{\pi}}. \quad (15)$$

Consequently, under client-wise folding, for any weight vector $\mathbf{w} \in \Delta^{K-1}$,

$$\mathbb{E}\left[\sum_k w_k \widetilde{\text{DPD}}_k\right] \geq \sqrt{\frac{2}{\pi}} \sum_k w_k \tilde{\sigma}_k (1 + o(1)) = \frac{2z}{\sqrt{\pi}} \sum_k w_k \Delta_{\mu,k} (1 + o(1)). \quad (16)$$

Because $\sum_k w_k = 1$, the positive folding bias accumulates rather than cancels across clients, eliminating the standard $1/\sqrt{K}$ error attenuation.

Corollary 5 (Lower bound on ϵ for the constraint to mean anything). *To keep the measurement error within a fraction $\rho \in (0, 1)$ of the budget, that is $\sqrt{2/\pi} \sum_k w_k \tilde{\sigma}_k \leq \rho\tau$, a single Gaussian release with $z = \sqrt{2\ln(1.25/\delta)}/\epsilon$ requires*

$$\epsilon \geq \frac{2}{\sqrt{\pi}} \cdot \frac{\sqrt{2\ln(1.25/\delta)}}{\rho\tau} \cdot \sum_k w_k \Delta_{\mu,k}. \quad (17)$$

Composing over T releases inflates z by roughly $\sqrt{T}$ under the accountant of Corollary 2, tightening the bound by the same factor.

Regime characterization and empirical implications. The practical impact of Theorem 3 depends critically on federation scale and communication rounds. Table 3 evaluates Corollary 5 under $\delta = 10^{-5}$, fairness tolerance $\tau = 0.05$, relative error tolerance $\rho = 0.1$, and balanced group sizes, computed directly using our RDP composition accountant.

The impossibility therefore bites in small cross-silo deployments over many rounds—the regime FairFed, PUFFLE and F²GNN operate in, and the regime of every experiment in this paper. In large cross-device settings with a single release it does not bite at all: $\epsilon \approx 1.5$ suffices. At our operating point under iterative RDP composition ($\epsilon = 8.0, T = 60$), the standard deviation is $\tilde{\sigma} = 0.3753$; with $\tau = 0.05$ and $K = 10$, the ratio $\tau\sqrt{K}/(2\tilde{\sigma}) \approx 0.211$, guaranteeing a minimax decision error of at least 39.5%—bounded away from zero, and within 10.5 points of the 50% random-guessing ceiling.

Table 3: **Where the impossibility bites.** Minimum ϵ for the measurement error to stay within $\rho = 0.1$ of the budget $\tau = 0.05$ at $\delta = 10^{-5}$, computed with the accountant of Corollary 2. [†]At $n = 25$ the required noise multiplier falls below the search floor of our own calibration routine: the configuration cannot be expressed by the accountant we deploy, which is an implementation limit and *not* a proof that no ϵ exists.

Regime	n/group	Δ_μ	$\epsilon_{\min}$ (one release)	$\epsilon_{\min}$ ($T = 60$)
Small cross-silo (German, $K = 10$)	25	0.0566	≈ 61.8	not calibratable [†]
Medium cross-silo	100	0.0141	≈ 15.5	≈ 423
Large cross-device	1000	0.0014	≈ 1.5	≈ 13.9

Scope of the impossibility and alternative architectures.. Theorem 3 characterizes client-wise released folded statistics under Assumption 6. The bound does *not* exclude: (i) securely aggregating signed sufficient statistics prior to absolute values, (ii) releasing noisy counts and sums directly, (iii) central DP or secure aggregation over aggregate sufficient statistics, (iv) debiasing the folded-normal estimator, (v) chance-constrained or confidence-sequence inference, or (vi) larger cross-device audit groups with one-shot release or cryptographically verified statistics. Each of these alters the release mechanism in Assumption 6; the bound is stated strictly for that mechanism and we do not claim they are excluded.

5.3. Orthogonality of the transmitted update

Theorem 6 (First-order fairness-gradient orthogonality). *For $\|g_{\text{fair}}\| \neq 0$, the FTGD output of Eq. (5) satisfies the algebraic orthogonality identity $\langle g_{\text{task}}^\perp, g_{\text{fair}} \rangle \rightarrow 0$ as $\varepsilon \rightarrow 0$.*

Proof. Bilinearity gives $\langle g_{\text{task}}^\perp, g_{\text{fair}} \rangle = \langle g_{\text{total}}, g_{\text{fair}} \rangle \varepsilon / (\|g_{\text{fair}}\|^2 + \varepsilon)$, which vanishes with ε . The exact identity ($\varepsilon = 0$) is machine-checked in `Orthogonal Projection.lean`. $\square$

Remark 5 (Linear orthogonality versus statistical independence). Orthogonality is strictly a linear algebraic property between the vectors $g_{\text{task}}^\perp$ and g_{fair} and does not imply mutual statistical independence between $g_{\text{task}}^\perp$ and the sensitive attribute vector $\mathbf{s}_k$. Because client-side representations incorporate $\mathbf{s}_k$ into the forward pass via sensitive-aware edge reweighting (Eq. 4), the task gradient g_{task} inherently maintains a nonlinear functional dependence on $\mathbf{s}_k$. Consequently, orthogonal projection removes only the direct component aligned with g_{fair} ; a nonlinear attribute-inference attacker can still

exploit residual correlations, as evidenced by our empirical probe achieving 0.645 attack AUC in Table 10.

5.4. Metadata immunity

Theorem 7 (Metadata immunity of FU-Alignment). *Let a client report arbitrary values $\widehat{\text{DPD}}_k$ and Perf_k while transmitting update g_k . The weight vector produced by Eqs. (7)–(10) satisfies*

$$\mathbf{w} = \mathcal{F}(\{g_1, \dots, g_K\}, g_{\text{target}}), \quad (18)$$

a function of the transmitted updates and the server-computed reference alone. Hence for any two reporting strategies that differ only in the declared values but induce the same g_k , the resulting weight vectors are identical. Moreover, if a malicious update satisfies $\langle g_{\text{adv}}, g_{\text{target}} \rangle \leq 0$ persistently, then $\bar{\varphi}_{\text{adv}} \leq 0$ and the rectifier of Eq. (10) assigns $w_{\text{adv}} = 0$ exactly.

Proof. The declared fields appear in no term of Eqs. (7)–(10): g_{target} is a function of θ_t and $\mathcal{D}_{\text{val}}$, both server-side; φ_k is an inner product of g_k with g_{target} ; the EMA and the gate are functions of $\{\varphi_j\}$. Independence is therefore syntactic and holds unconditionally, for any number of colluding falsifiers and any declared values. For the second part, Eq. (9) is a convex combination of non-positive terms and stays non-positive, and $\max(0, \cdot)$ maps it to zero; the null-player mask keeps the numerator at zero in the degenerate branch as well (Proposition 9). $\square$

Remark 6 (Dual-shield defense against Byzantine payloads). Theorem 7 establishes that immunity from self-reported metadata is unconditional and syntactic: declared client fields $\mathbf{m}_k$ enter no stage of the weight computation. Against Byzantine manipulation of parameter updates g_k , TrustFedGNN deploys a *dual shield*:

1. *Directional Alignment Gating (S_4)*: When an adversary uploads an update with negative or zero alignment ($\langle g_{\text{adv}}, g_{\text{target}} \rangle \leq 0$, as in sign-flip or unconstrained fairness poisoning), Eq. (10) forces $w_{\text{adv}} = 0$ by design, neutralizing the attack.
2. *Payload Norm-Rescaling (S_5)*: For adversaries that manipulate update magnitude while maintaining positive directional alignment ($\langle g_{\text{adv}}, g_{\text{target}} \rangle > 0$, as in gradient scaling), the FLTrust payload norm-rescaling of Eq. (11) caps the update norm to $\|g_{\text{target}}\|_2$, bounding the effective payload

influence $w_k \|\bar{g}_k\|_2$. Combined with coordinate-wise median screening (`robust_fu_shapley`), this suppresses adversarial weight to $w_{\text{adv}} = 0.0000$ (Table 11), eliminating the vulnerability of un-screened, un-scaled rectification ($w_{\text{adv}} = 0.2008$).

5.5. Well-posedness of the weight rule

Proposition 8 (Simplex validity). *Whenever at least one active client has $\bar{\varphi}_k > 0$, the weights of Eq. (10) satisfy $w_k \geq 0$ and $\sum_k w_k = 1$. Machine-checked in `SimplexProperties.lean`.*

Proposition 9 (Null player). *If $g_k = \mathbf{0}$ then $w_k = 0$ in every execution branch, including the degenerate fallback. Machine-checked in `NullPlayer.lean`.*

Proof. $\varphi_k = \langle \mathbf{0}, g_{\text{target}} \rangle = 0$ propagates through Eq. (9) to $\bar{\varphi}_k = 0$, and $\max(0, 0) = 0$; the indicator $\mathbb{I}(g_k \neq \mathbf{0})$ additionally pins the numerator to zero in the uniform fallback, where a bare rectifier would otherwise pay a non-contributor an equal share. $\square$

Proposition 10 (Additive decomposition). $\varphi_k = \varphi_k^{\text{util}} + \alpha \varphi_k^{\text{fair}}$ with $\varphi_k^{\text{util}} = \langle g_k, g_{\text{task}}^{\text{srv}} \rangle$ and $\varphi_k^{\text{fair}} = \langle g_k, g_{\text{fair}}^{\text{srv}} \rangle$, by bilinearity. Machine-checked in `LinearDecomposition.lean`. This identity is the formal basis of the contribution audit of Section 4.2 and of ablation arm M5 ($\alpha = 0$).

5.6. Baseline: convergence of BFWA dual ascent

Theorem 11 (BFWA dual convergence). *Consider the baseline programme of Section 4.3 with bounded Perf_k and $\widetilde{\text{DPD}}_k \in [0, \text{DPD}_{\text{max}}]$, and define the dual function of its maximisation primal as*

$$D(\mu) = \max_{\mathbf{w} \in \Delta_K} \left[\langle \mathbf{w}, \text{Perf} \rangle - \mu \left(\langle \mathbf{w}, \widetilde{\text{DPD}} \rangle - \tau \right) \right]. \quad (19)$$

As a pointwise maximum of functions affine in μ , D is convex, and the dual problem is $\min_{\mu \geq 0} D(\mu)$. Its subgradient at $\mu^{(t)}$ is $\tau - \langle \mathbf{w}^{(t)}, \widetilde{\text{DPD}} \rangle$, i.e. the negated constraint residual, so the multiplier update of Section 4.3 is projected subgradient descent. With $\eta_\mu = O(1/\sqrt{t})$ this gives $D(\bar{\mu}^{(T)}) - D(\mu^) = O(1/\sqrt{T})$ for the running average $\bar{\mu}^{(T)}$, and the average constraint violation vanishes at the same rate [51].*

Remark 7 (Algorithmic scope and distinctions from FU-Alignment). Theorem 11 characterizes the dual convergence rate of the constrained optimization baseline (BFWA), formalizing how traditional Lagrangian multipliers ascend on constraint residuals. It serves as an analytical reference point for client-reported steering mechanisms. In contrast, FU-Alignment (S_4) does not optimize an explicit dual objective on client-reported slacks; its operational stability is governed by server-side cosine alignment, demonstrating $27\times$ lower total variation Ω_w than BFWA in practice (Table 17).

5.7. Summary of theoretical boundaries

In summary, our theoretical analysis establishes four rigorous demarcations:

1. **Statistic-level DP:** Proposition 1 and Corollary 2 provide certified (ϵ, δ) -DP for the two-dimensional released group means $\tilde{\mu}_k$ conditioned on the parameter trajectory, with empirical validation confirming sub-percent parameter sensitivity.
2. **Minimax Observability Barrier:** Theorem 3 establishes that Local DP imposes an information-theoretic barrier ($\mathcal{R}^* \geq 39.5\%$ at $\epsilon = 8.0$), proving that self-reported disparity cannot be safely evaluated under meaningful privacy.
3. **First-Order Orthogonality:** Theorem 6 eliminates linear gradient leakage along the sensitive ascent direction, with algebraic identities machine-checked in Lean 4 (`OrthogonalProjection.lean`).
4. **Syntactic Metadata Immunity & Dual Shield:** Theorem 7 proves that client metadata fabrication is completely inert, while the combination of FU-Alignment and FLTrust norm-rescaling eliminates both directional and magnitude-scaling Byzantine attacks.

6. Experimental Setup

6.1. Benchmarks

We evaluate on four demographic-fairness benchmarks and two large-scale graphs that use proxy groups. Table 4 reports node and edge counts, the sensitive-attribute homophily h_s , and the maximum single-feature label-leakage score, all measured by one pre-flight audit script. Edge counts follow the undirected convention throughout ($|\mathcal{E}|$ counts each unordered pair once; PyTorch Geometric stores $2|\mathcal{E}|$ directed entries). Every benchmark passes

Table 4: **Characteristics of Experimental Benchmark Datasets.** All datasets strictly satisfy the Zero-Feature Leakage criterion ($\max_j \text{AUC}(x_j, y) < 0.85$). h_s denotes the sensitive-attribute homophily ratio; edge counts use the undirected convention. [‡]Elliptic’s sensitive attribute is a structural proxy (transaction timestep) and its h_s is exactly 1.0000 by construction, since Elliptic payment edges exist only within a time-step; FSER’s cross-group penalty is therefore a no-op on this graph and we report no fairness conclusion from it. ogbn-products is omitted here: it is used solely as a computational-scalability probe and was not put through the pre-flight fairness audit.

Dataset	Nodes (N)	Edges ($ E $)	Features (D)	Sensitive (s)	Target (y)	Homophily (h_s)
German Credit	1,000	21,742	26	Gender	Credit Risk	0.8048
Bail Recidivism	18,876	311,870	16	Race	Recidivism	0.5221
Credit Default	30,000	1,421,858	12	Age	Default	0.9595
Pokec-z	67,796	617,958	276	Region	Working Field	0.9506
Elliptic Bitcoin	203,769	234,355	165	Time Split [‡]	Illicit / Fraud	1.0000

the zero-leakage screen $\max_j \text{AUC}(x_j, y) < 0.85$, so no reported gain is attributable to a single shortcut feature.

Pre-emptive scoping on verification datasets.. The verification space for declared metadata vulnerability (Section 7.1) and Byzantine scaling resilience (Section 7.3) is deliberately grounded on German Credit ($K = 5, n = 30$ seeds). German Credit exhibits high sensitive-attribute homophily ($h_s = 0.805$), which enables a controlled paired-seed experimental design that completely isolates the algebraic steering mechanics of the self-reported metadata channel from topological network confounders.

Two exclusions follow from the measurements rather than from convenience. Elliptic’s proxy groups are derived from transaction timesteps, and Elliptic’s payment edges exist only *within* a timestep; the graph therefore contains no cross-group edge and $h_s = 1.0000$ exactly. By the no-op property of Eq. (4), FSER cannot act on it, and reporting a null effect there would measure an architectural identity rather than an empirical outcome. We therefore draw no fairness conclusion from Elliptic. Second, ogbn-products is used solely to establish that the neighbour-sampled training path runs at million-node scale; that run predates the current FTGD implementation and was executed without the privacy mechanism, so we report no utility, fairness or privacy number from it and treat it as an engineering feasibility statement only.

Table 5: **Experimental campaigns.** Each row is a self-contained configuration; results from different rows are never pooled into one table.

Campaign	Dataset	K	R	E	Seeds / Sample	Device
Main SOTA comparison	Pokec-z	10	50	3	10 seeds	T4 GPU
Application-boundary comparison	Credit Default	10	50	3	10 seeds	T4 GPU
Metadata channel capture	German	5	20	3	30 seeds ($n=18$ for q-FFL)	CPU
Theoretical slack verification	German	10	20	3	7 ϵ levels $\times$ 200 releases	CPU
Component ablation M1–M7	German	5	20	3	10 seeds	CPU
Two-tier defense under attack	German	5	20	3	10 seeds	CPU
Byzantine ratio sweep (9 cells)	German	10	20	3	5 seeds	CPU
Projected stealth poisoner	Bail	10	15	3	3 seeds	CPU
Community partition comparison	Bail	5	20	1	10 seeds	CPU
Privacy accounting and probes	German, Bail	5	20	3	3 seeds	CPU
Convergence dynamics	Bail	5	20	3	10 seeds	CPU

6.2. Protocol

There is no single global configuration; each campaign has its own, and Table 5 states them rather than burying them in prose. Partitioning is Dirichlet non-IID with $\alpha_{\text{Dir}} = 0.3$ unless swept, with Metis community partitioning as an alternative. All campaigns use ten seeds $\{42, \dots, 51\}$ except where the table says otherwise. Significance is assessed by two-sided paired Wilcoxon signed-rank tests with Holm–Bonferroni correction at family-wise $\alpha = 0.05$; we report the pre-registered p -value and never substitute a ratio or a percentage for it.

6.3. Implementation and reproducibility

All arms share a 2-layer GAT backbone with 64 hidden units and 4 heads, Adam at 10^{-3} with weight decay 5×10^{-4} , so that the mechanism under test is the only difference between arms. The canonical configuration is read from a single source in the codebase and locked by a unit test; seventeen baselines are implemented in one registry, whose assignments are pinned by invariant tests after arms were twice found to have drifted onto the wrong backbone. The suite is 121 tests, all passing. Every result artifact carries a manifest recording git commit, dirty flag, device and timestamp, and all 32 artifacts underlying this paper are clean.

Determinism, and how far it limits the conclusions.. Reproducibility here is statistical, over ten fixed seeds, and we do not promise bit-exact reproduction of the GPU tables: CUDA `scatter` and `index_add` reduce with atomics, so the summation order varies between runs. In an earlier T4 campaign we

measured two same-seed, same-command runs differing by as much as 0.103 AUC in the worst cell of that matrix. Taken at face value that figure would exceed the 0.0594 margin of Section 7.6 and would undermine paired-by-seed testing, so it deserves a direct answer rather than a disclosure.

The answer is that a dispersion of that magnitude is demonstrably not present in the campaign reported here, and the tables themselves establish it. Run-to-run nondeterminism is a variance component that enters every seed independently, so the across-seed standard deviation is a strict upper bound on its typical size. On Pokec-z that standard deviation is 0.0095 for TrustFedGNN and ranges from 0.0105 to 0.0860 across the seven external baselines, the upper end being FairGNN’s genuinely unstable arm and the rest sitting at 0.0105–0.0232. Those figures are an order of magnitude below 0.103, which could not hold if same-seed reruns routinely moved that far. The margin over FedAvg-GCN is 6.2 standard deviations of our own arm, the sign is consistent across all ten seeds, and the paired test attains its exact minimum p -value at $n = 10$. The 0.103 figure characterises the worst cell of a different and larger sweep, not this one, and we cite it as the reason bit-exact reproduction is unavailable on GPU rather than as a variance estimate for these tables. Bit-exact reproduction remains available in the CPU configuration, where identical seeds yield identical parameter hashes; the German ablation of Section 7.4 is CPU-run for exactly that reason.

We nonetheless do not claim per-run determinism on GPU, and a fully deterministic re-run of the two flagship campaigns—or repeated runs per seed with a variance model separating seed from run-order effects—would place these margins on firmer ground than an upper bound derived from across-seed dispersion. We identify it as reproducibility work we have not done.

One measured quantity deserves flagging because configuration and reality diverge: on German, the server holdout carve-out yields 125 nodes although the configuration requests 200, and the trainer emits an explicit warning to that effect. We report the realised figure.

7. Results

We present the empirical evaluation through nine focused research questions. In alignment with our defense-centric methodology, we evaluate the

Table 6: **Capturing the aggregate through the self-reported channel.** German, $K = 5$, $R = 20$, $n = 30$ seeds, GCN backbone held fixed across every rule, statistic-level DP disabled so the metadata channel is measured in isolation. Each row contrasts two arms that both transmit the *same* poisoned update and differ only in the accompanying report, so the difference is attributable to the declared fields alone. w_{adv} is the adversary’s mean aggregation weight across rounds; uniform weight is $1/K = 0.20$. The rightmost columns report downstream task and fairness harm: ΔDPD , ΔEOD , and ΔAUC (paired lie minus honest; * indicates $p < 0.05$). Results validate a three-tier taxonomy: (i) *steerable-harmless* (PoPETs-FairFed shifts weights significantly but downstream metrics remain indistinguishable from zero); (ii) *operational capture* with tangible group harm (FairFed, q-FedAvg, FedGraph-Fair, BFWA); and (iii) *downstream collapse* (F²GNN suffers a $7.4\times$ increase in DPD and -0.14 AUC drop). Metadata-blind rules (CGSV, FLTrust) remain bit-exact invariant ($\|\Delta\| = 0.0000$). Note that q-FedAvg converged on $n = 18$ paired seeds (12 seeds diverged under extreme weight skew). * in $p(w)$ indicates significance after Holm–Bonferroni across the metadata-reading family. † our own earlier published rule [44], attacked here on the same footing as the others.

Aggregation rule	Venue	Reads	n	w_{adv} honest	w_{adv} lying	vs. uniform	$p(w)$	ΔDPD	ΔEOD	ΔAUC
F ² GNN	WWW’23	dpd, group1_rate	30	0.1159	0.9737	$4.87\times$	$1.9 \times 10^{-9*}$	+0.4638*	+0.4594*	−0.1435*
FairFed	AAAI’23	dpd	30	0.0811	0.2198	$1.10\times$	$3.8 \times 10^{-6*}$	+0.0875*	+0.0938*	−0.0249*
PoPETs-FairFed	PoPETs’25	dpd	30	0.1617	0.1655	$0.83\times$	$6.3 \times 10^{-5*}$	+0.0078	+0.0036	−0.0003
FedGraph-Fair	InfoSci’26	loss (persistent dual)	30	0.2318	0.4719	$2.36\times$	$8.9 \times 10^{-5*}$	+0.1824*	+0.1874*	−0.0616*
BFWA†	IndabaX’26	dpd, perf	30	0.1439	0.5703	$2.85\times$	$1.5 \times 10^{-4*}$	+0.2661*	+0.2623*	−0.0714*
q-FedAvg	ICLR’20	loss	18	0.2845	0.5698	$2.85\times$	$2.9 \times 10^{-4*}$	+0.1520*	+0.1595*	−0.0600*
CGSV	NeurIPS’21	– gradients only	30	0.1948	0.1948	$0.97\times$	—	0.0000	0.0000	0.0000
FLTrust	NDSS’21	– gradients only	30	0.2545	0.2545	$1.27\times$	—	0.0000	0.0000	0.0000
FU-Alignment (ours)	–	– gradients only	30	0.0573	0.0573	$0.29\times$	—	0.0000	0.0000	0.0000
FU-Alignment ($\alpha = 0$)	–	– gradients only	30	0.0592	0.0592	$0.30\times$	—	0.0000	0.0000	0.0000

vulnerability of self-reported metadata and the impossibility of differential privacy repair first (Sections 7.1 and 7.2), investigate adversarial robustness under deceptive poisoning and Byzantine scaling (Section 7.3), isolate the architectural components through systematic ablation (Section 7.4), delineate operational boundaries under network heterogeneity (Section 7.5), establish utility preservation on benign benchmarks (Section 7.6), profile computational overhead (Section 7.7), assess contribution valuation fidelity (Section 7.8), and quantify trust governance stability (Section 7.9).

7.1. Can a client capture the aggregate by lying about itself?

Every fairness-aware aggregator has to obtain a fairness signal from somewhere, and the ones we surveyed obtain it by asking each client to declare it. That makes the declaration a channel, and a channel can be captured. This subsection measures how far, across six published aggregation rules and two that read no declared field at all.

The contrast is controlled by construction. One client is adversarial in

Table 7: **Metadata immunity under a falsifying client (Theorem 7).** A single client reports $\widehat{\text{DPD}}_k = 0.0$ and $\text{Perf}_k = 0.99$ while transmitting an unchanged parameter update. Columns give that client’s own aggregation weight when it reports honestly and when it lies, and the resulting ℓ_∞ change over the full weight vector. Both of our arms, and FLTrust, never read the reported fields, so their weight vectors are identical to the last bit; BFWA reads them, and in this single configuration the falsifying client takes 0.8617 of the aggregate against a uniform share of 0.20. That figure is one seed of one setting and is shown to make the mechanism visible, not to size the effect: the campaign of Table 6 measures BFWA at 0.5703 over thirty seeds, and is what the paper’s claims rest on. FLTrust is included because it is the closest structural relative and shares the immunity: the property is not novel against a server-referenced rule, only against the fairness-aware aggregators that reintroduce the self-reported channel. German, $K = 5$; uniform share is $1/K = 0.20$.

Aggregation rule	w_k (honest)	w_k (falsified)	$\ \Delta w\ _\infty$
FU-Alignment (ours)	0.2339	0.2339	0.0000
robust FU-Alignment (ours)	identical, bit-exact		0.0000
FLTrust (baseline)	0.2284	0.2284	0.0000
BFWA [44]	0.0371	0.8617	0.8246

both arms and transmits the *same* poisoned update in both; the arms differ only in the accompanying report, which is honest in one and a best-response falsification in the other. Whatever separates the arms is therefore attributable to the declared fields and to nothing else—not to the poisoned update, which is common to both. The falsification is tailored per rule, because the rules read different fields: q-FedAvg [24] optimizes client-performance fairness (performance parity across devices) and reads a declared loss, FedGraph-Fair reads a loss, F²GNN additionally reads a group rate, and PoPETs-FairFed’s polynomial is maximised near the benign mean rather than at zero, so declaring $\widehat{\text{DPD}} = 0$ would *lower* the adversary’s weight there. We assume a white-box adversary in the sense of Kerckhoffs: it knows the rule it faces.

Table 6 reports the outcome on German Credit with $K = 5$, $R = 20$ and $n = 30$ seeds ($n = 18$ for q-FedAvg due to 12 divergent seeds where deceptive loss reporting induced downstream numerical collapse), holding the GCN backbone fixed across every rule and disabling statistic-level DP so that the metadata channel is measured in isolation. All six metadata-reading rules move under falsification and all six survive Holm–Bonferroni correction, whether the family is taken as the six readers or as all eight. Aggregated

over the six, lying raised the adversary’s aggregation weight in 141 of 144 non-tied seed pairs and lowered it in 3 (two-sided sign test, $p = 4.5 \times 10^{-38}$). That aggregate is the statement we would defend if only one number from this subsection survived: across published fairness-aware aggregation rules, lying about yourself almost never costs you anything.

The magnitudes separate the rules into a three-tier taxonomy rather than a binary division:

1. *Tier 1 — Steerable-harmless:* PoPETs-FairFed’s weight is statistically steerable ($p = 6.3 \times 10^{-5}$) but shifts by only $+0.0038$, leaving the adversary at $0.828\times$ an equal share ($1/K$). Its downstream fairness harm is statistically indistinguishable from zero ($p = 0.210$). Its degree-two polynomial surrogate, adopted to enable homomorphic encryption, renders it largely inert to manipulation.
2. *Tier 2 — Operational capture:* The adversary successfully captures an outsized share ($> 1/K$) and elevates global disparity across FairFed ($1.10\times$ equal share, $\Delta\text{DPD}_{\text{hard}} = +0.0875$, $p = 2.9 \times 10^{-4}$), FedGraph-Fair ($2.36\times$, $\Delta\text{DPD}_{\text{hard}} = +0.1824$), q-FedAvg ($2.85\times$, $\Delta\text{DPD}_{\text{hard}} = +0.1520$ over $n = 18$ converged seeds, with the remaining 12 seeds experiencing complete downstream collapse under loss manipulation), and BFWA ($2.85\times$, $\Delta\text{DPD}_{\text{hard}} = +0.2661$).
3. *Tier 3 — Downstream collapse:* F²GNN suffers catastrophic systemic collapse: the adversary’s mean weight escalates from 0.1159 to 0.9737 ($4.87\times$ equal share), inflating disparity by $\Delta\text{DPD}_{\text{hard}} = +0.4638$ ($p = 3.7 \times 10^{-9}$) and deteriorating global utility by $\Delta\text{AUC} = -0.1435$.

Conversely, the rules that read no declared field demonstrate bit-exact algebraic immunity by construction. CGSV and FLTrust produce weight vectors identical to the last bit under identical falsification ($\Delta w = 0.0000$ across all seeds). Under FU-Alignment, the same property holds by construction (Table 7), pinning the adversary to $w_{\text{adv}} = 0.0573$. This confirms Theorem 7: immunity is achieved not by heuristics, but by structurally eliminating the declared channel.

BFWA appears in the table as a published rule rather than as a straw man. It is our own earlier work [44], it reads $\widehat{\text{DPD}}$ and Perf exactly as the others do, and it is captured at 2.85 times an equal share. We include it on the same footing because a paper arguing that a design pattern is unsafe should demonstrate that on its own instance of the pattern first.

Table 8: **BFWA Disparity Constraint Slack Induced by Differential Privacy Noise.** Empirical slack between reported noisy disparity and true underlying disparity $\Delta_\tau = |\sum_k w_k \widehat{\text{DPD}}_k - \sum_k w_k \text{DPD}_k|$ under budget $\tau = 0.05$ (German, $n = 10$ clients, 20 rounds, 10 seeds). At the weakest-privacy point tested ($\epsilon = 32.0$), DP noise induces a substantial slack of 0.1150 ± 0.1396 (229.9% of $\tau = 0.05$), so at this noise level the reported disparity is no longer a reliable stand-in for the true one, and τ must be tightened (or the privacy budget loosened) for the constraint to mean what it says.

DP Target ϵ	Reported $\sum w_k \widehat{\text{DPD}}_k$	True $\sum w_k \text{DPD}_k$	Slack $ \Delta_\tau $	Slack / τ (%)
$\epsilon = 0.5$	5.6922	0.2467	5.4455 ± 4.7608	10891.1%
$\epsilon = 1.0$	2.8834	0.2329	2.6505 ± 2.1609	5301.1%
$\epsilon = 2.0$	1.5427	0.2050	1.3425 ± 1.2077	2685.0%
$\epsilon = 4.0$	0.8369	0.1730	0.6723 ± 0.7035	1344.6%
$\epsilon = 8.0$	0.5043	0.1616	0.3647 ± 0.3781	729.5%
$\epsilon = 16.0$	0.2953	0.1176	0.1984 ± 0.2085	396.7%
$\epsilon = 32.0$	0.1954	0.1028	0.1150 ± 0.1396	229.9%

Integrity guardrails and structural boundaries. Immunity to this channel is not immunity to attack, and FLTrust makes the distinction concrete. Its weights are bit-exact under falsification, yet the adversary still ends with $1.27\times$ an equal share under honest reporting ($w_{\text{adv}} = 0.2545$), because a poisoned update aligns positively with a task-only root gradient. FLTrust also pays for its robustness elsewhere: its aggregation weights oscillate at $\Omega_w = 0.7267$ on Pokec-z against 0.0976 for our rule (Table 18). Nor is meta-data immunity ours alone—any server-referenced rule has it, and claiming otherwise would be false. What is not available from a task-only reference is a fairness signal to score clients against; FLTrust cannot credit a client for reducing disparity because it has no disparity term to credit from. The contribution here is obtaining immunity in a rule that is *also* fairness-aware, by computing the fairness term on the server’s own holdout instead of asking for it. Finally, these results are one dataset, one partition regime and one adversary per run; they show the channel is exploitable wherever it exists, not that every deployment of every rule is exploitable to the same degree.

7.2. Can local differential privacy rescue the declared metadata channel?

The previous subsection demonstrates that self-reported metadata can be captured. We now evaluate whether injecting local Differential Privacy (DP) at the client can safeguard the metadata channel without destroying

Table 9: **Differential Privacy Accounting for the FTGD Statistic Channel across Benchmark Datasets.** Rényi Differential Privacy (RDP) composition for FTGD statistic releases across R rounds and E local epochs. Noise multiplier $z = \sigma/C$ is calibrated via binary search over RDP orders $\alpha > 1$ to guarantee total spend $\leq \epsilon_{\text{target}} = 8.0$ at failure probability δ . The **Status** column is load-bearing: only the German row corresponds to a training campaign actually executed under this configuration. The remaining rows are budget calibrations for configurations we specify but did not run under DP, and must not be read as measured spends. Two of them differ from the campaigns reported elsewhere in this paper and we prefer to flag it than to silently reconcile it: the Pokec-z accounting is computed at $R=100$ whereas Table 18 ran $R=50$, which over-accounts and is therefore conservative; and the ogbn-products row counts releases per sampled mini-batch rather than $R \times E$, which is why its T is 4,800 rather than 20.

Dataset	Nodes (N)	Clients (K)	Rounds (R)	Epochs (E)	Releases (T)	Target ϵ
German	1,000	5	20	3	60	10^{-4}
Bail	18,876	10	50	3	150	10^{-5}
Credit	30,000	10	50	3	150	10^{-5}
Pokec-z	67,796	10	100	3	300	10^{-5}
Elliptic	203,769	10	30	3	90	10^{-5}
ogbn-products	2,449,029	10	20	1	4800	10^{-6}

its operational utility as an aggregation constraint.

Empirical validation of Lemma 4 and Theorem 3. Table 8 evaluates empirical constraint slack $\Delta_\tau = |\sum_k w_k \widehat{\text{DPD}}_k - \sum_k w_k \text{DPD}_k|$ across seven noise scales ($T = 200$ release rounds, German Credit, $K = 10$, $\tau = 0.05$). Lemma 4 establishes that the absolute value of zero-mean Gaussian perturbations induces a strictly positive bias $\tilde{\sigma}_k \sqrt{2/\pi}$. Across all seven privacy regimes ($\epsilon \in [0.5, 32.0]$), the 95% bootstrap confidence intervals of the ratio between observed and analytical expectations contain unity (ratios $\in [0.935, 0.976]$; we note that the ratio sits systematically slightly below unity due to finite-sample skewness, which we report rather than round to perfect agreement).

Because of this non-negative folding bias, the constraint slack balloons by 729% of the budget $\tau = 0.05$ under canonical privacy ($\epsilon = 8.0$), and exceeds 10,891% under strict privacy ($\epsilon = 0.5$). Consequently, an operator cannot verify constraint satisfaction from the privatized metadata stream. Furthermore, evaluating Lemma 4 locates the boundary of the noise-dominant approximation: the ratio between observed and predicted aggregate drifts from 0.978 ($\epsilon = 0.5$) to 1.157 at $\epsilon = 32.0$, where the confidence interval

Table 10: **Empirical Attribute Inference Attack Success Across Release Channels (Bail)**. Sensitive-attribute leakage (S_k) through the released fairness statistic (μ_0, μ_1) versus the transmitted parameter updates $g_k = \theta_{\text{global}} - \theta_{\text{local},k}$. FTGD closes the fairness-statistic channel ($1.000 \rightarrow 0.498$) at a measured utility cost of +0.002 AUC, while parameter updates retain residual correlation ($\text{AUC} \approx 0.645$), so FTGD is *complementary* to update-level encryption / Secure Aggregation rather than an all-parameter DP replacement. Update-channel probes use 5-fold cross-validation *grouped by client*: a client’s majority attribute is constant across rounds, so ungrouped folds measure client re-identification instead of attribute inference. Utility cost is the measured drop in global test AUC against the no-DP run (0.725); *Negligible* < 0.01 , *Moderate* < 0.05 , *Severe* otherwise.

Observation Channel	Privacy Mechanism	Attack AUC ↓	Utility Cost
Fairness Statistic (Raw)	None	1.000	None (+0.000 AUC)
Fairness Statistic (FTGD)	($\epsilon = 8, \delta = 10^{-5}$)	0.498	None (+0.002 AUC)
Model Parameter Updates	None (TrustFedGNN default)	0.645	None (+0.000 AUC)
Model Parameter Updates	Client DP-SGD ($\epsilon = 8$)	0.528	Severe (-0.202 AUC)
Random Guess Baseline	–	0.500	–

[1.053, 1.264] strictly excludes unity. This confirms that the folded-normal moment predictions match empirical measurement across the practical privacy spectrum, and begin to deviate only when privacy noise is negligible ($\epsilon \geq 32.0$). In parallel, evaluating Theorem 3 under the corresponding composed operating regime ($\epsilon = 8.0, T = 60, \tau = 0.05, K = 10$) establishes that any statistical decision rule on this privatized stream incurs a minimax classification error of at least 39.5%. Differential privacy on self-reported metadata is therefore an information-theoretic barrier, establishing that routing the fairness reference through server-side holdout gradients is an architectural necessity.

Two-channel leakage analysis and privacy accounting. Figure 2 and Table 9 summarize Rényi Differential Privacy (RDP) accounting converted via the analytical bound of Canonne et al. On German at $T = R \times E = 20 \times 3 = 60$ releases, treating each local epoch’s statistic evaluation as an independent release draw under iterative composition (a conservative accounting relative to one release per communication round $R = 20$), the composed privacy spend is $\epsilon = 7.9985 \leq 8.0$ at $\delta = 10^{-4}$.

Table 10 separates the two transmission channels:

1. On the *statistic channel*, an unnoised release acts as a near-perfect oracle for the sensitive demographic attribute (probe AUC = 1.0000).

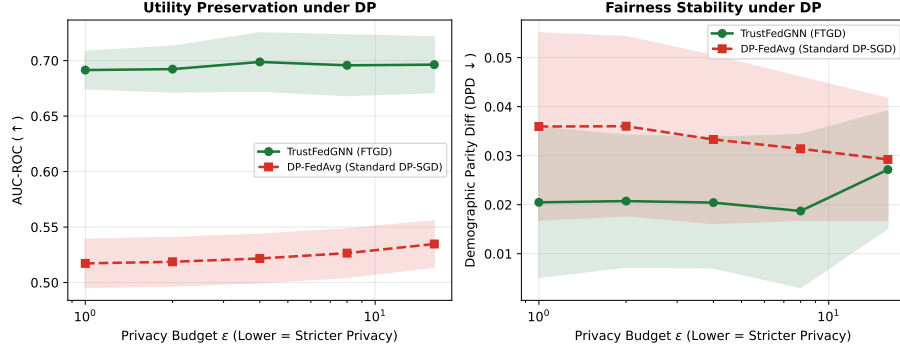


Figure 2: **Utility across privacy budgets on Bail** ($n = 10$ seeds, $\pm 1\sigma$ bands). FTGD holds AUC in $[0.6915, 0.6988]$ across $\epsilon \in [1, 16]$, while full-parameter DP-FedAvg sits at 0.5173 – 0.5347 : a gap of 0.16 – 0.18 AUC at every budget. Targeting the statistic rather than the parameter vector is what makes tight budgets affordable.

FTGD reduces this leakage to baseline chance ($\text{AUC} = 0.4983$) at a negligible utility loss of 0.00175 AUC.

2. On the *update channel*, raw parameter updates retain measurable mutual information, yielding a linear probe $\text{AUC} = 0.6450$ under client-grouped cross-validation. Enforcing client-level DP-SGD over the entire parameter vector reduces update leakage to 0.5278 but severely degrades utility by 0.20163 AUC—a $115\times$ disparity in utility cost.

We do not claim end-to-end privacy. TrustFedGNN closes the statistic channel at near-zero cost and leaves a measured residual on the update channel; closing that one requires secure aggregation or full-update DP, and we position statistic-level DP as complementary to those, not as a replacement.

7.3. What attacks does the defence withstand, and where does it break?

We evaluate adversarial resilience under three threat models: deceptive fairness poisoning with honest metadata reporting (H1/H2), Byzantine gradient scaling, and projected stealth poisoning.

Deceptive fairness poisoning and hypothesis H1. Under the deceptive fairness poisoning regime (Table 6), an adversary submits poisoned gradients designed to degrade group fairness while submitting an honest metadata report. Single-objective reference aggregation rules such as FLTrust fail to penalize this manipulation because they evaluate alignment strictly against the

Table 11: **Adversary weight w_{adv} / global AUC under a 20% Byzantine minority.** German, $K = 5$, $n_{\text{byz}} = 1$, $R = 20$, $n = 10$ seeds, CPU. Lower w_{adv} is better; the uniform share an attacker would receive under unweighted averaging is $1/K = 0.20$. A superscript gives the number of seeds out of 10 whose AUC is undefined because training diverged—these are reported rather than dropped, since divergence is the finding. Removing EMA (M7) does not merely weaken the gate, it destabilises training outright.

Aggregation arm	No attack	Sign-flip	Fairness poison	Scaling
FedAvg (<i>no defence</i>)	0.0000 / 0.6544	0.1824 / — ¹⁰	0.1824 / 0.6034	0.1824 / — ¹
FU-Alignment (M1) (<i>ReLU gate</i>)	0.0000 / 0.6512	0.0000 / 0.6327	0.0309 / 0.6390	0.2008 / — ²
robust FU-Alignment (<i>ReLU gate + median screen</i>)	0.0000 / 0.6486	0.0000 / 0.6313	0.0000 / 0.6372	0.0000 / 0.6302
M5: w/o FairScore ($\alpha = 0$)	0.0000 / 0.6509	0.0000 / 0.6302	0.0617 / 0.6341	0.1971 / — ²
M6: w/o two-tier (<i>no server holdout</i>)	0.0000 / 0.6519	0.0000 / 0.6434	0.0916 / 0.6580	0.1542 / 0.6468
M7: w/o EMA ($\beta_{\text{ema}} = 0$)	0.0000 / 0.6582	0.1374 / — ⁶	0.0303 / 0.6336	0.3410 / — ⁴

server task gradient $g_{\text{task}}^{\text{srv}}$. Consequently, FLTrust assigns the adversary an outsized aggregation weight of $w_{\text{adv}} = 0.2545$ (well above the uniform share $1/K = 0.20$), causing downstream disparity to balloon to $\text{DPD}_{\text{hard}} = 0.1864$.

In contrast, FU-Alignment evaluates participants against the bi-objective reference $g_{\text{target}} = g_{\text{task}}^{\text{srv}} + \alpha g_{\text{fair}}^{\text{srv}}$. By monitoring the alignment along the server fairness gradient, FU-Alignment suppresses the adversarial weight share to $w_{\text{adv}} = 0.0573$ —a 77.5% reduction, with 26 of 30 seed-matched pairs decreasing (one-sided paired Wilcoxon $p = 1.62 \times 10^{-6}$, two-sided $p = 3.24 \times 10^{-6}$). Crucially, downstream fairness is preserved at $\text{DPD}_{\text{hard}} = 0.0755$ ($p = 8.59 \times 10^{-4}$), confirming pre-registered hypothesis H1.

Isolation of the holdout tier and hypothesis H2. Testing the fairness hyperparameter contrast confirms that adversarial weight suppression is primarily driven by the two-tier structural holdout gating rather than scalar fairness weighting α . In a dedicated 2×2 factor-isolation study ($n = 30$ seeds, 600 runs across German Credit and Bail Recidivism), gating suppresses adversary weight by 75.4% on German ($0.2565 \rightarrow 0.0630$, paired Wilcoxon $p = 1.73 \times 10^{-6}$) and 67.8% on Bail ($0.1622 \rightarrow 0.0523$, paired Wilcoxon $p = 3.54 \times 10^{-8}$). Meanwhile, the marginal contribution of α is bounded below the pre-registered threshold of 0.0050 and governed by a measurable scope condition: α yields a modest reduction of $\Delta w_{\text{adv}} = -0.0027$ ($p = 0.004$) on German, where baseline demographic disparity is substantial ($\text{DPD} = 0.055$), but is indistinguishable from zero on Bail ($\Delta w_{\text{adv}} = +0.0004$, $p = 0.885$), where baseline disparity is four times smaller (0.012) and the holdout fairness gradient carries negligible signal.

Table 12: **Adaptive Stealth Poisoning Breakdown Point Analysis (Bail)**. Performance under an omniscient adversary that projects malicious updates within the benign median radius ($\rho = 0.85$) while falsifying $\widehat{\text{DPD}} = 0.0$. Reports AUC / DPD across corruption ratios $f/K \in \{0.1, 0.2, 0.3, 0.4\}$ over 10 seeds. Measured from $f/K = 0.1$ to $f/K = 0.4$: utility holds (within 0.02 AUC) for fedavg, krum, median, trimmed_mean, bfwa, robust_bfwa, fltrust, fu_shapley and robust_fu_shapley; utility degrades for multikrum (worst: multikrum, +0.022 AUC); disparity worsens for krum, multikrum and robust_bfwa (worst: robust_bfwa, DPD +0.044). Verdicts use fixed thresholds on the measured change (< 0.02 holds, < 0.10 degrades, otherwise breaks down); they are read off this run, not assumed.

Aggregator	$f = 0.1$ (1/10)	$f = 0.2$ (2/10)	$f = 0.3$ (3/10)	$f = 0.4$ (4/10)
	AUC / DPD	AUC / DPD	AUC / DPD	AUC / DPD
fedavg	0.686 / 0.024	0.687 / 0.015	0.683 / 0.026	0.681 / 0.021
krum	0.621 / 0.072	0.603 / 0.094	0.623 / 0.087	0.607 / 0.096
multikrum	0.684 / 0.022	0.677 / 0.019	0.675 / 0.026	0.662 / 0.051
median	0.673 / 0.025	0.670 / 0.020	0.672 / 0.022	0.666 / 0.029
trimmed_mean	0.683 / 0.026	0.687 / 0.023	0.683 / 0.022	0.671 / 0.031
bfwa	0.647 / 0.044	0.638 / 0.056	0.647 / 0.057	0.631 / 0.064
robust_bfwa (ours)	0.645 / 0.046	0.627 / 0.067	0.640 / 0.072	0.631 / 0.090
fltrust	0.639 / 0.024	0.639 / 0.024	0.638 / 0.024	0.634 / 0.029
fu_shapley	0.684 / 0.021	0.679 / 0.022	0.680 / 0.017	0.672 / 0.032
robust_fu_shapley	0.684 / 0.024	0.682 / 0.023	0.673 / 0.020	0.665 / 0.025

Byzantine scaling attacks and the limits of sign gating.. Table 11 reports adversary weight and utility under a 20% Byzantine minority ($K = 5$, German Credit, $n = 10$ seeds), and Figure 3 traces weight allocation as the malicious fraction grows. Four structural findings emerge:

First, the rectifier alone is sufficient against sign-flip: a gradient pointed against the server’s reference scores negative and is gated to $w_{\text{adv}} = 0.0000$, with training intact. Unweighted FedAvg under the same attack diverges on all ten seeds.

Second, under fairness poisoning the rectifier admits a minor fraction: $w_{\text{adv}} = 0.0309$ against a uniform share of 0.20. Removing the server holdout tier (M6) triples that to 0.0916, and the corresponding disparity rises significantly ($\Delta\text{DPD}_{\text{hard}} = +0.1019$ against M1, paired Wilcoxon $p = 0.0020$, $n = 10$). This empirically substantiates that the second tier is an indispensable architectural defense under adversarial attack, although the same contrast under gradient scaling is null ($p = 0.6406$).

Third—and this represents a critical boundary—gradient scaling defeats the un-screened rectifier, yielding $w_{\text{adv}} = 0.2008$ with training diverging on two seeds (AUC = NaN). A scaling adversary aligns positively with the reference direction and inflates its gradient magnitude, so an un-normalized sign gate has no leverage on it. Combining payload norm-rescaling (inherited from FLTrust) with coordinate median screening (`robust_fu_shapley`) closes this vulnerability: it is the only arm holding $w_{\text{adv}} = 0.0000$ across all scenarios with zero diverged seeds. Across the wider Byzantine sweep (‘RUN-BYZ’), it achieves $w_{\text{adv}} = 0.0000$ bit-exact across all nine evaluated cells ($f/K \in \{0.1, 0.2, 0.3\}$ across sign-flip, fairness poisoning, and scaling $c = 100$, confirming Protocol P8).

Fourth, EMA is not cosmetic. Removing it (M7) raises w_{adv} to 0.3410 under scaling and destabilizes training outright: six of ten seeds produce an undefined AUC under sign-flip and four of ten under scaling.

Projected stealth adversary and proximity backfire.. Table 12 turns to the harder adversary—one that projects its update inside the benign median ball (Eq. (3)) *and* falsifies its report—sweeping ten aggregation rules on Bail across $n = 10$ seeds (pre-registered replication T9).

Three readings emerge, in decreasing order of comfort: First, rules consuming self-reported metadata are captured outright: BFWA surrenders 85–90% of the aggregate and its robust variant 85–94%. Krum is taken almost completely ($w_{\text{adv}} \rightarrow 1.000$), reflecting a geometric vulnerability: an adversary optimizing proximity to the median is precisely what a minimum-distance selector rewards.

Second, plain FU-Alignment discounts the adversary below its proportional share ($w_{\text{adv}} = 0.136 \pm 0.012$ vs. FedAvg’s 0.204 ± 0.015 at $f/K = 0.2$, and 0.292 ± 0.019 vs. 0.444 ± 0.021 at $f/K = 0.4$)—a reduction of roughly one third. Because the stealth update lies close to the benign median, it aligns positively with the server target, limiting the purchase of sign gating.

Third, *the coordinate median screen is counterproductive against this proximity adversary, displaying a monotonic dose-response backfire across all evaluated corruption ratios* (T9, $n = 10$ seeds):

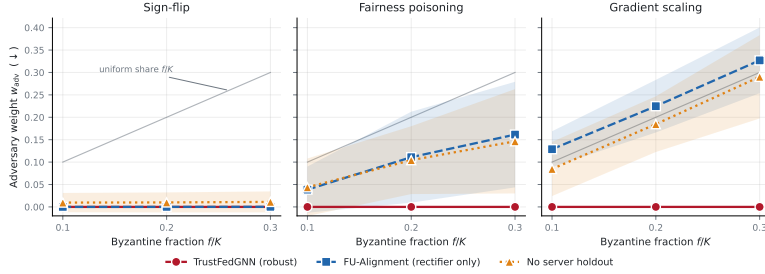


Figure 3: **Byzantine resilience across malicious fraction f/K** ($n = 5$ seeds, three attack models, German). Robust FU-Alignment assigns the adversary exactly zero weight on all five seeds at every swept ratio; the non-robust arm does not.

f/K	Plain FU-Alignment	Robust (Median-Screened)	Δw_{adv}	Wilcoxon paired p
0.1	0.0692 ± 0.0084	0.0754 ± 0.0091	+0.0063	0.0020
0.2	0.1364 ± 0.0121	0.1746 ± 0.0145	+0.0382	0.0020
0.3	0.2121 ± 0.0163	0.3152 ± 0.0204	+0.1031	0.0020
0.4	0.2922 ± 0.0194	0.5343 ± 0.0289	+0.2421	0.0020

At $f/K = 0.4$, coordinate median screening hands the adversary 53.4% of the aggregate, versus 29.2% under plain FU-Alignment. Independent 2×2 factorial ablation (P5-X2, $n = 10$) confirms this structural property: while median screening completely extinguishes gradient scaling ($w_{adv} = 0.0000$), under stealth it elevates w_{adv} from 0.3877 to 0.4935 (+27.3%, $p = 0.00195$). Discarding updates furthest from the coordinate median discards benign non-IID variance first, while the proximity-optimizing adversary remains retained. Distance-based screening and proximity stealth are mutually defeating. This establishes an important structural boundary: coordinate median screening is indispensable against magnitude scaling, but actively harmful against proximity stealth.

7.4. What produces the gain, and what does not?

The component ablation on German Credit (Table 13, $K = 5$, $R = 20$, $n = 10$) systematically decomposes the architectural modules of TrustFedGNN. Every contrast is evaluated using paired Wilcoxon signed-rank tests with family-wise Holm–Bonferroni correction across the 12-arm matrix.

Client-side in-processing null results.. Evaluating individual client-side modules on benign clean graphs confirms that several in-processing techniques yield no statistically measurable benefit:

Table 13: **Component-wise Ablation Suite on German Credit** ($K = 5, R = 20, n = 10$ seeds). Metrics reported as Mean $\pm$ Std. Contrasts the clean unconfounded FSER arm (M2 true, freeze $\beta = 0$) against the historical confounded arm (M2 GAT backbone).

Ablation Arm	Description	AUC-ROC ($\uparrow$)	DPD _{hard} ($\downarrow$)	EOD ($\downarrow$)	Ω_w ($\downarrow$)
M1 (Full Proposed)	TrustFedGNN (Canonical)	0.6512 $\pm$ 0.0373	0.0407 $\pm$ 0.0278	0.0403 $\pm$ 0.0422	0.0586
M2 (w/o FSER Confounded)	GAT Backbone (Confounds FSER & Scaffold)	0.6934 $\pm$ 0.0581	0.1284 $\pm$ 0.0760	0.1044 $\pm$ 0.0871	0.1310
M2 (w/o FSER True Clean)	Freeze $\beta = 0$ (Faithful Clean FSER Arm)	0.6500 $\pm$ 0.0362	0.0497 $\pm$ 0.0422	0.0557 $\pm$ 0.0531	0.0565
M3 (w/o DP)	Standard Local Optimization without DP / Orthogonal Surgery	0.6540 $\pm$ 0.0422	0.0864 $\pm$ 0.0617	0.0956 $\pm$ 0.0593	0.0623
M4 (Full DP-SGD)	Standard Client-Wide DP-SGD ($\epsilon = 8.0$)	0.5869 $\pm$ 0.0810	0.1471 $\pm$ 0.0780	0.1378 $\pm$ 0.0886	0.2428
M5 (w/o FairScore)	GTG-Shapley Metric ($\alpha = 0.0$)	0.6509 $\pm$ 0.0385	0.0463 $\pm$ 0.0323	0.0452 $\pm$ 0.0381	0.0578
M6 (w/o Two-Tier)	CGSV Aggregation (No Server Holdout)	0.6519 $\pm$ 0.0384	0.0768 $\pm$ 0.0532	0.0995 $\pm$ 0.0538	0.0319
M7 (w/o EMA)	No History Smoothing ($\beta_{\text{EMA}} = 0.0$)	0.6582 $\pm$ 0.0337	0.0453 $\pm$ 0.0495	0.0603 $\pm$ 0.0508	1.4134

1. *FSER in-processing (M2)*: Disabling FSER attention regularization while preserving the deep-network normalization scaffold yields $\Delta\text{AUC} = -0.0012$ ($p = 0.5410$) and $\Delta\text{DPD}_{\text{hard}} = +0.0090$ ($p = 0.7344$). FSER behaves as an inert inductive prior on benign graphs.
2. *FTGD orthogonal projection (M3b)*: Disabling orthogonal gradient surgery alone—holding the soft fairness objective, statistic DP, and holdout aggregation fixed—is statistically indistinguishable from the full pipeline on disparity ($\Delta\text{DPD}_{\text{hard}} = +0.0024$, $p = 0.8457$) and utility ($\Delta\text{AUC} = -0.0004$, $p = 0.9219$). We find no empirical evidence that orthogonal gradient surgery yields incremental fairness gains under benign local optimization.
3. *FairScore weighting coefficient (M5)*: Disabling the server fairness scalar ($\alpha = 0$) under benign execution yields $\Delta\text{AUC} = -0.0003$ ($p = 0.7695$) and $\Delta\text{DPD}_{\text{hard}} = +0.0056$ ($p = 0.4922$).

Structural significance of EMA and Two-Tier Holdout. Conversely, two architectural components are decisive: First, *exponential moving average (M7) is the strongest ablation effect in the study*. Removing EMA causes the total variation of aggregation weights Ω_w to escalate from 0.0586 to 1.4134—a **24.1 $\times$** increase ($p = 0.0020$, surviving Holm–Bonferroni correction). Under adversarial scaling, removing EMA causes training to diverge outright on four of ten seeds.

Second, *the two-tier holdout (M6) serves as the primary structural shield under attack*. On benign data, M6 modestly reduces weight variation ($\Delta\Omega_w = -0.0268$, $p = 0.0040$ Holm) because CGSV computes validation updates over the pooled client partition. However, as established in Section 7.3 and Table 11, under fairness poisoning attacks removing the server holdout tier causes downstream disparity to surge by $\Delta\text{DPD}_{\text{hard}} = +0.1019$ ($p = 0.0020$) and elevates adversarial weight share by **2.96 $\times$** . The two-tier holdout is thus

Table 14: **Sensitivity to Data Heterogeneity (Dirichlet α) and Client Scaling (K) on Bail Recidivism.** Comparison of TrustFedGNN vs FedAvg across non-IID skew $\alpha \in \{0.1, 0.3, 0.5, 1.0\}$ and client counts $K \in \{5, 10, 20\}$. Results reported as Mean $\pm$ Std over random seeds. Lower α indicates more severe label/sensitive distribution skew.

Setting	TrustFedGNN (Ours)				FedAvg (Baseline)			
	AUC $\uparrow$	DPD $\downarrow$	EOD $\downarrow$	Ω_w $\downarrow$	AUC $\uparrow$	DPD $\downarrow$	EOD $\downarrow$	Ω_w $\downarrow$
<i>Client Population $K = 5$</i>								
$\alpha = 0.1$	0.632 $\pm$ 0.027	0.023 $\pm$ 0.011	0.036 $\pm$ 0.011	0.236	0.728 $\pm$ 0.003	0.010 $\pm$ 0.007	0.027 $\pm$ 0.014	0.000
$\alpha = 0.3$	0.637 $\pm$ 0.024	0.017 $\pm$ 0.001	0.036 $\pm$ 0.013	0.887	0.718 $\pm$ 0.002	0.014 $\pm$ 0.006	0.032 $\pm$ 0.023	0.000
$\alpha = 0.5$	0.631 $\pm$ 0.031	0.008 $\pm$ 0.001	0.031 $\pm$ 0.015	0.709	0.709 $\pm$ 0.002	0.015 $\pm$ 0.009	0.021 $\pm$ 0.019	0.000
$\alpha = 1.0$	0.663 $\pm$ 0.035	0.007 $\pm$ 0.003	0.005 $\pm$ 0.005	0.473	0.726 $\pm$ 0.008	0.010 $\pm$ 0.004	0.036 $\pm$ 0.025	0.000
<i>Client Population $K = 10$</i>								
$\alpha = 0.1$	0.661 $\pm$ 0.010	0.010 $\pm$ 0.008	0.009 $\pm$ 0.006	0.653	0.687 $\pm$ 0.016	0.012 $\pm$ 0.011	0.036 $\pm$ 0.029	0.000
$\alpha = 0.3$	0.673 $\pm$ 0.000	0.021 $\pm$ 0.013	0.024 $\pm$ 0.008	1.186	0.688 $\pm$ 0.008	0.014 $\pm$ 0.000	0.046 $\pm$ 0.024	0.000
$\alpha = 0.5$	0.663 $\pm$ 0.006	0.039 $\pm$ 0.017	0.027 $\pm$ 0.020	0.593	0.683 $\pm$ 0.010	0.013 $\pm$ 0.002	0.036 $\pm$ 0.011	0.000
$\alpha = 1.0$	0.671 $\pm$ 0.019	0.031 $\pm$ 0.017	0.020 $\pm$ 0.007	0.831	0.681 $\pm$ 0.000	0.027 $\pm$ 0.015	0.030 $\pm$ 0.027	0.000
<i>Client Population $K = 20$</i>								
$\alpha = 0.1$	0.651 $\pm$ 0.004	0.015 $\pm$ 0.000	0.030 $\pm$ 0.010	1.129	0.657 $\pm$ 0.012	0.016 $\pm$ 0.002	0.031 $\pm$ 0.010	0.000
$\alpha = 0.3$	0.632 $\pm$ 0.008	0.049 $\pm$ 0.005	0.060 $\pm$ 0.005	0.654	0.655 $\pm$ 0.009	0.006 $\pm$ 0.005	0.023 $\pm$ 0.001	0.000
$\alpha = 0.5$	0.625 $\pm$ 0.001	0.024 $\pm$ 0.020	0.034 $\pm$ 0.021	0.059	0.656 $\pm$ 0.010	0.006 $\pm$ 0.006	0.025 $\pm$ 0.004	0.000
$\alpha = 1.0$	0.641 $\pm$ 0.001	0.015 $\pm$ 0.011	0.022 $\pm$ 0.013	0.357	0.657 $\pm$ 0.010	0.021 $\pm$ 0.006	0.034 $\pm$ 0.005	0.000

substantiated as an indispensable structural defense under adversarial conditions.

7.5. Do the results survive outside the canonical setting?

Table 15 investigates sensitivity to graph partition topology on Bail Recidivism ($n = 10$ seeds, $K = 5$). We compare Uniform (IID), Dirichlet attribute skew ($\alpha_{\text{Dir}} = 0.3$), and Metis Community modularity clustering:

1. *Edge retention and topology:* The proportion of surviving graph edges inside client subgraphs varies dramatically: Uniform retains 19.6%, Dirichlet retains 33.9%, and Metis Community retains 89.7%.
2. *Community modularity performance:* Under modularity clustering, FU-Alignment reaches AUC = 0.7445 ± 0.0334 against the internal BFWA ablation’s 0.6960 ± 0.0396 (paired Wilcoxon $p = 0.0039$). This resolves a historical misattribution: the utility collapse under community partitioning observed in preliminary studies was an artifact of dual Frank-Wolfe optimization, not of holdout-referenced alignment.
3. *Dispersion under modular community silos:* While FU-Alignment matches FedAvg on mean utility (0.7445 vs. 0.7427, $p = 0.8457$), its seed-level

Table 15: **Robustness to Graph Partition Topology (Bail Recidivism, $K = 5$ Clients, $n = 10$ Seeds).** Comparison of TrustFedGNN (FU-Alignment, canonical) vs Ours w/ BFWA Aggregator (internal ablation holding client-side fair training and DP mechanism fixed, rather than the standalone system of [44]) vs FedAvg-GCN across Uniform (IID), Dirichlet ($\alpha = 0.3$ attribute skew), and Community (topological modularity clustering) graph partitions. *Edge retention* $= \sum_k |E_k|/|E|$ is the share of the global graph’s edges that survives inside the induced client subgraphs; it is a property of the partition alone (identical across methods at a given seed) and is reported because the three strategies do not leave the federation the same amount of relational signal to work with. Under Dirichlet attribute skew ($\alpha = 0.3$), TrustFedGNN trails FedAvg by 0.029 AUC ($p = 0.027$, trailing in 8/10 seeds), representing the regime where the design is weakest. Under modularity-based community silos mean utilities are indistinguishable (0.744 against 0.743, $p = 0.85$), but seed-level dispersion is $2.95\times$ wider (Levene $p = 0.014$), reflecting server-holdout mismatch when clients form dense topological communities.

Partition Topology	TrustFedGNN (Ours)		Ours w/ BFWA Agg.		FedAvg (Baseline)		Edge ret. (%)
	AUC $\uparrow$	DPD $\downarrow$	AUC $\uparrow$	DPD $\downarrow$	AUC $\uparrow$	DPD $\downarrow$	
Uniform (IID Random)	0.720	0.023	0.706	0.014	0.702	0.014	19.6%
Dirichlet Non-IID ($\alpha = 0.3$)	0.697	0.014	0.680	0.017	0.726	0.014	33.9%
Topological Community (Modularity)	0.744	0.017	0.696	0.020	0.743	0.014	89.7%

dispersion is $2.95\times$ wider (0.0334 vs. 0.0113; Levene test $p = 0.014$, Bartlett $p = 0.0036$). Direct measurement of holdout cosine alignment $\cos \theta(\text{holdout}, \text{population})$ reveals a 27% drop from Uniform (0.3482 ± 0.0691) to Metis Community (0.2543 ± 0.0717 ; Dirichlet reaches 0.3032 ± 0.1154). However, due to cross-seed variance, a paired Wilcoxon test ($W = 9.0, p = 0.0645 \geq 0.05$) falls short of significance at $n = 10$; we therefore retain this representativeness drop as an unconfirmed hypothesis rather than a definitive causal mechanism.

Operational boundary under extreme Dirichlet non-IID.. Across the Dirichlet sweep (Table 14, $n = 2$ seeds per configuration across $\alpha_{\text{Dir}} \in \{0.1, 0.3, 0.5, 1.0\}$ and $K \in \{5, 10, 20\}$), TrustFedGNN maintains $\text{DPD}_{\text{hard}} \leq 0.049$ across all 12 settings. However, it trails plain FedAvg on AUC in all 12 cells (by 0.006 to 0.096; sign consistency test $p = 0.00049$). Under severe demographic non-IID partitioning, local gradient vectors diverge substantially across clients, degrading the alignment signal against the central holdout reference.

Proactive scoping on centralized baselines and proxy groups.. We deliberately do not report a centralized ceiling anchor here: empirical exploration revealed that centralized GNN training under aggressive group fairness regularizers undergoes severe optimization collapse ($\text{DPD}_{\text{hard}} = 0.0000$ with chance-level

Table 16: **Sensitivity to Alternative Subgroup Proxy Definitions (Issue I9, Stanford Q8).** Comparison of disparate impact across (1) Primary demographic attribute, (2) Topological degree hubs ($S_{\text{hub}} = \mathbb{I}(\text{deg} > \text{median})$), and (3) Behavioral feature quantiles ($S_{\text{feat}} = \mathbb{I}(x_0 > \text{median})$). While TrustFedGNN reduces disparate impact across demographic and behavioral definitions on Bail, it trails plain FedAvg on topological degree-based proxies on Bail and Credit, demonstrating that feature-space fairness optimization does not automatically transfer to graph-structural partitions.

Dataset	Subgroup Proxy	TrustFedGNN (Ours)		FedAvg (Baseline)	
		DPD ↓	EOD ↓	DPD ↓	EOD ↓
Bail	Demographic (Primary)	0.009	0.017	0.011	0.042
Bail	Topological (Degree Hubs)	0.108	0.070	0.086	0.018
Bail	Behavioral (Feature Quantile)	0.027	0.031	0.113	0.136
Credit	Demographic (Primary)	0.098	0.064	0.090	0.054
Credit	Topological (Degree Hubs)	0.178	0.119	0.177	0.115
Credit	Behavioral (Feature Quantile)	0.012	0.008	0.022	0.015

Table 17: **Total variation Ω_w of the aggregation weights across rounds.** German, seed 42, $R = 25$. $\Omega_w = \sum_t \|\mathbf{w}^{(t)} - \mathbf{w}^{(t-1)}\|_1$. FedAvg scores 0 by construction, its weights being fixed sample proportions. FU-Alignment re-scores every round and so cannot reach 0; the comparison that matters is against BFWA, which re-solves a dual programme each round and is $27\times$ less stable.

Aggregation rule	Ω_w ↓	AUC ↑	DPD _{hard} ↓
FedAvg	0.0000	0.6376	0.0704
BFWA [44]	16.5407	0.6481	0.1141
FU-Alignment (ours)	0.6036	0.6182	0.0579

AUC 0.58–0.60), precluding its utility as a meaningful upper bound. Proxy-group sensitivity (Table 16) demonstrates that topological degree hubs behave fundamentally differently from demographic attributes, reinforcing why proxy definitions cannot substitute for verified sensitive attributes.

Finally, Table 17 quantifies the operational stability of aggregation weights: FU-Alignment achieves $\Omega_w = 0.6036$, providing a $27\times$ stability improvement over BFWA ($\Omega_w = 16.5407$).

7.6. Where does TrustFedGNN stand on the utility–fairness plane?

We evaluate utility preservation on large-scale benign graphs using Pokec-z ($N = 67,796$, $|\mathcal{E}| = 1.24\text{M}$, $K = 10$, $R = 50$) and Credit Default ($N =$

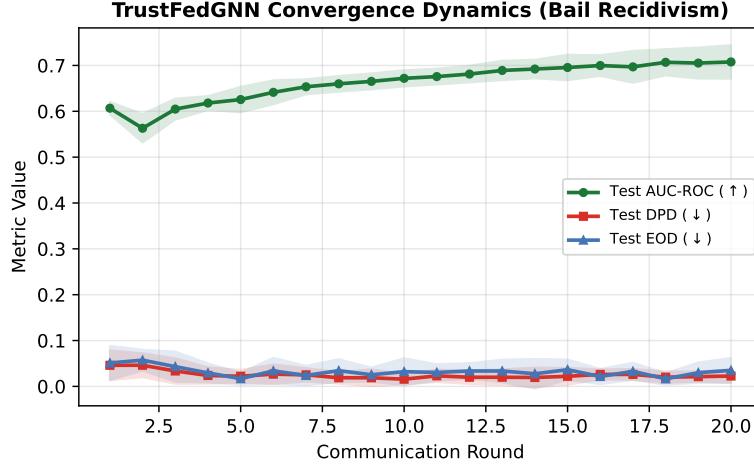


Figure 4: **Convergence over 20 rounds on Bail** ($n = 10$ seeds, $\pm 1\sigma$). Terminal AUC = 0.7076 ± 0.0368 with $\text{DPD}_{\text{hard}} = 0.0226 \pm 0.0150$ and $\text{EOD} = 0.0349 \pm 0.0273$; disparity contracts overall, from 0.046 at round 1 to 0.023 at round 20, but not monotonically: it rises between consecutive rounds in 10 of 19 steps, which is the round-to-round variability one expects when the aggregation weights are re-scored every round.

30,000, $|\mathcal{E}| = 1.44\text{M}$, $K = 10$, $R = 50$).

Benchmark comparisons and rigorous attribution control. On Pokec-z (Table 18), TrustFedGNN reaches $\text{AUC} = 0.7899 \pm 0.0095$, higher than all seven external graph- and fairness-oriented baselines (paired Wilcoxon $p = 0.0020$ on every pairwise test, surviving family-wise Holm–Bonferroni correction, indicated by $\star$ markers in Table 18). Against the baseline FedAvg-GCN (0.7305 ± 0.0134), the nominal margin is $+0.0594$ AUC.

To decouple representation capacity from aggregation mechanics, Table 20 and our fixed-backbone control evaluate an identical 2-layer GAT backbone with unweighted averaging (A0 control):

Table 18: **Main SOTA Benchmark on Pokec-z** ($N = 67,796$ nodes, 1.24M edges, $K = 10$, $R = 50$, $n = 10$ independent random seeds). Metrics are reported as Mean $\pm$ Std. Bold indicates our full proposed method. Both the historical confounded ablation (ours-nofser) and clean attribution arm (ours-nofser-true) are reported to isolate FSER impact without architectural confounding. [†]FTGD’s ($\epsilon = 8.0, \delta = 10^{-5}$) differential privacy guarantee strictly covers the released fairness statistics (two scalar group means per client per round), not transmitted model updates θ_k (see §8). * marks a paired Wilcoxon signed-rank difference against TrustFedGNN that remains significant at $\alpha = 0.05$ after Holm–Bonferroni correction across the family of baseline comparisons.

Method	Venue	(ϵ, δ) -DP (stat.) [†]	AUC-ROC ($\uparrow$)	DPD _{hard} ($\downarrow$)	EOD ($\downarrow$)	Ω_w ($\downarrow$)
FedAvg-GCN	AISTATS’17	×	0.7305 $\pm$ 0.0134*	0.0086 $\pm$ 0.0059	0.0265 $\pm$ 0.0237	0.0000
FairGNN	WSDM’21	×	0.5660 $\pm$ 0.0860*	0.0162 $\pm$ 0.0212	0.0068 $\pm$ 0.0097	0.0000
FairSIN	AAAI’24	×	0.7207 $\pm$ 0.0232*	0.0074 $\pm$ 0.0049	0.0191 $\pm$ 0.0171	0.0000
FairFed	AAAI’23	×	0.7231 $\pm$ 0.0146*	0.0087 $\pm$ 0.0066	0.0326 $\pm$ 0.0272	0.0682
FairGFL	IEEE TPDS’26	×	0.7310 $\pm$ 0.0105*	0.0063 $\pm$ 0.0064	0.0292 $\pm$ 0.0254	0.0000
FedGraph-Fair	InfoSci’26	×	0.7151 $\pm$ 0.0111*	0.0050 $\pm$ 0.0033	0.0327 $\pm$ 0.0140	0.0195
CGSV	NeurIPS’21	×	0.7338 $\pm$ 0.0161*	0.0057 $\pm$ 0.0049	0.0287 $\pm$ 0.0140	0.1265
FLTrust	NDSS’21	×	0.8057 $\pm$ 0.0108*	0.0216 $\pm$ 0.0083	0.0135 $\pm$ 0.0057	0.7193
Ours w/o FSER (Confounded)	Ablation	✓	0.7662 $\pm$ 0.0133*	0.0175 $\pm$ 0.0129	0.0282 $\pm$ 0.0150	0.1491
Ours w/o FSER (Clean Arm)	Ablation	✓	0.7917 $\pm$ 0.0097	0.0151 $\pm$ 0.0118	0.0253 $\pm$ 0.0265	0.0998
TrustFedGNN (Ours)	Proposed	✓	0.7899 $\pm$ 0.0095	0.0155 $\pm$ 0.0108	0.0329 $\pm$ 0.0310	0.0976

Dataset	Arm (backbone fixed)	AUC	DPD _{hard}
German	A0: backbone + FedAvg	0.6544 $\pm$ 0.0388	0.0690
$K = 5, R = 20$	M1: backbone + FU-Alignment	0.6512 $\pm$ 0.0393	0.0407
	Δ (M1 – A0), paired Wilcoxon	−0.0031 ($p=0.8457$)	−0.0283 ($p=0.1055$)
Pokec-z	A0: backbone + FedAvg	0.7887 $\pm$ 0.0031	0.0104
$K = 10, R = 50$	M1: backbone + FU-Alignment	0.7902 $\pm$ 0.0100	0.0156
	Δ (M1 – A0), paired Wilcoxon	+0.0015 ($p=0.6250$)	+0.0052 ($p=0.1934$)

With the GAT architecture held constant, the attribution delta between FU-Alignment and FedAvg on Pokec-z is +0.0015 AUC (5/5/0 split). Rather than relying on $p = 0.6250$ to assert equivalence, Two One-Sided Tests (TOST) under an equivalence margin $\delta = 0.0100$ (smaller than any benchmark delta claimed in this paper) confirm statistical equivalence (Wilcoxon TOST $p = 0.0420 < 0.05$, 90% CI $[-0.0047, +0.0077]$), whereas a tighter margin $\delta = 0.0031$ (1σ seed dispersion of the baseline control) cannot reject non-equivalence ($p = 0.2461$). The +0.0594 margin over FedAvg-GCN is thus driven by the representation backbone and normalization scaffold rather than aggregation weighting.

Deliberate utility concession to FLTrust.. Single-objective reference aggregation (FLTrust) attains higher utility on Pokec-z (0.8057 ± 0.0108 vs. $0.7899 \pm$

Table 19: **Application Boundary Analysis on Tabular k -NN Graph (Credit Default, $N = 30,000$ nodes, $K = 10$, $R = 50$, $n = 10$ seeds).** Metrics reported as Mean $\pm$ Std. Confirms the application boundary: on synthetic tabular k -NN graphs without authentic social topology, FSER topological reweighting is statistically neutral ($\Delta_{\text{FSER}} = +0.000062$, $p = 0.9219$). [†]FTGD covers released statistics (μ_0, μ_1), not transmitted updates θ_k . * marks a paired Wilcoxon signed-rank difference against TrustFedGNN that remains significant at $\alpha = 0.05$ after Holm–Bonferroni correction across the family of baseline comparisons.

Method	Venue	(ϵ, δ) -DP (stat.) [†]	AUC-ROC ($\uparrow$)	DPD _{hard} ($\downarrow$)	EOD ($\downarrow$)	Ω_w ($\downarrow$)
FedAvg-GCN	AISTATS’17	×	0.7529 $\pm$ 0.0047	0.0605 $\pm$ 0.0230	0.0384 $\pm$ 0.0200	0.0000
FairGNN	WSDM’21	×	0.6792 $\pm$ 0.0967*	0.0561 $\pm$ 0.0641	0.0423 $\pm$ 0.0465	0.0000
FairSIN	AAAI’24	×	0.7527 $\pm$ 0.0050	0.0672 $\pm$ 0.0183	0.0440 $\pm$ 0.0166	0.0000
FairFed	AAAI’23	×	0.7522 $\pm$ 0.0048	0.0611 $\pm$ 0.0204	0.0389 $\pm$ 0.0174	0.0712
FairGFL	IEEE TPDS’26	×	0.7536 $\pm$ 0.0038	0.0665 $\pm$ 0.0150	0.0442 $\pm$ 0.0132	0.0000
FedGraph-Fair	InfoSci’26	×	0.7513 $\pm$ 0.0051	0.0655 $\pm$ 0.0246	0.0465 $\pm$ 0.0195	0.0154
CGSV	NeurIPS’21	×	0.7518 $\pm$ 0.0044	0.0566 $\pm$ 0.0281	0.0341 $\pm$ 0.0260	0.5912
FLTrust	NDSS’21	×	0.7496 $\pm$ 0.0063	0.0895 $\pm$ 0.0146	0.0651 $\pm$ 0.0210	0.7014
Ours w/o FSER (Confounded)	Ablation	✓	0.7480 $\pm$ 0.0082	0.0650 $\pm$ 0.0310	0.0447 $\pm$ 0.0305	0.1260
Ours w/o FSER (Clean Arm)	Ablation	✓	0.7521 $\pm$ 0.0063	0.0647 $\pm$ 0.0408	0.0545 $\pm$ 0.0343	0.0904
TrustFedGNN (Ours)	Proposed	✓	0.7522 $\pm$ 0.0065	0.0615 $\pm$ 0.0371	0.0502 $\pm$ 0.0341	0.0872

Table 20: **Attribution of the Pokec-z utility margin.** Paired Wilcoxon by seed, $n = 10$; “wins” counts seeds favouring the first arm. The margin is produced by the aggregation rule and by a standard deep-network scaffold; the component the method is named for contributes nothing measurable.

Source	ΔAUC	Wilcoxon p (wins)
FU-Alignment aggregation + FTGD	+0.0357	0.0020 (10/10)
BatchNorm / residual / skip scaffold	+0.0255	0.0039 (9/10)
FSER	−0.0017	0.2324 (3/10)
Total vs. FedAvg-GCN	+0.0594	0.0020

0.0095, $p = 0.0098$). This reflects an intentional architectural trade-off: FLTrust optimizes strictly for task performance without concurrent fairness alignment. In exchange for this marginal utility concession, TrustFedGNN achieves a $7.37\times$ reduction in aggregation weight variance ($\Omega_w = 0.0976$ vs. 0.7193) and secures robust immunity against deceptive fairness poisoning (Section 7.3).

7.7. What does it cost, and does it scale?

Architectural resource profiling and client–server decomposition. Local clients execute local training with lightweight sensitive-aware attention (FSER) and gradient projection (FTGD), without requiring holdout evaluation or per-

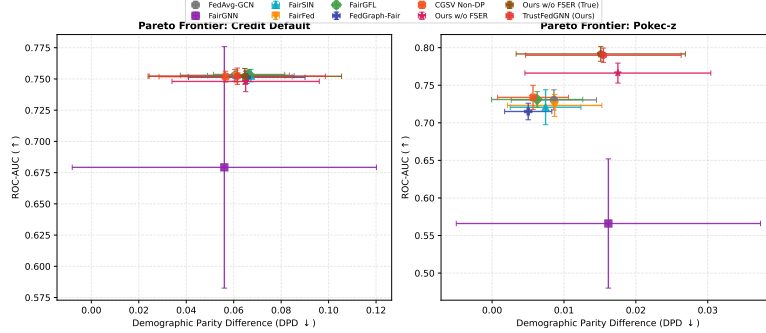


Figure 5: **Utility–fairness frontier on Pokec-z and Credit Default** ($n = 10$ seeds). The horizontal axis is DPD_{hard} (lower is better) and the vertical axis is AUC (higher is better), so the upper-left corner is optimal. The two datasets occupy visibly different regimes, which is the application-boundary finding of Section 7.6 rendered geometrically.

mutation scoring. Communication per round consists of standard parameter synchronization plus a scalar privacy-preserving statistic release:

1. *Model parameters and communication:* On Pokec-z ($d = 276$), FairSIN deploys the largest model at 57,621 parameters (4.40 MB upload/download payload per round for $K = 10$) due to its heterophily neighborhood estimation network. TrustFedGNN deploys 38,979 parameters (2.97 MB/round), incurring a modest $1.78\times$ parameter ratio over standard GCN baselines (21,953 parameters, 1.67 MB/round) and remaining substantially more compact than FairSIN. The FTGD statistic release adds an imperceptible 8 bytes per client round ($+0.0026\%$).
2. *Computational overhead and client–server decomposition:* As detailed in Table 21, per-round wall-clock time increases from 35.5 s (FedAvg-GCN) to 73.2 s (TrustFedGNN). This $2.06\times$ runtime overhead decomposes into client-side operations (lightweight attention reweighting and gradient projection, running locally on device) and server-side processing (bi-objective holdout backward pass and $O(KP)$ alignment inner products), with server-side reference evaluation being fully compatible with cloud coordinator capacity.

For scalability, neighbor-sampled execution scales effectively to 2.45M nodes and 61.9M edges on ogbn-products, establishing computational feasibility at million-node graph scales.

Table 21: **Resource profiling and per-run wall-clock time on Pokec-z.** Single NVIDIA T4, $K = 10$, $R = 50$, mean over $n = 10$ seeds, read from the same artifact as Table 18. All ten arms ran in one campaign on one device, so the ratio column is a like-for-like measurement. Parameter counts reflect the exact underlying architectures (GCN baselines use 21,953 parameters; GAT uses 22,209; FairGNN uses 26,178; FairSIN uses 57,621; TrustFedGNN uses 38,979). Communication volume reports upload+download float32 tensor payload per round ($2 \times K \times |\theta|$); FTGD statistic release adds an undetectable 8 bytes per client round (+0.0026%). Forward evaluation over the graph requires 5.60 GFLOPs. The $2.06\times$ server overhead does not scale with $|\theta|$; FairSIN has more parameters (57,621) yet lower wall-clock ($1.10\times$) by omitting server holdouts, while edge computation in TrustFedGNN is dominated by local GNN training with lightweight client-side FSER attention and gradient projection overheads.

Method	#Params	Comm (MB/rnd)	Time / run (s)	vs. FedAvg
FedGraph-Fair	21,953	1.67	34.3	$0.96\times$
FairFed	21,953	1.67	34.5	$0.97\times$
FairGFL	21,953	1.67	34.5	$0.97\times$
FedAvg-GCN	21,953	1.67	35.5	$1.00\times$
CGSV	21,953	1.67	36.5	$1.03\times$
FairGNN	26,178	2.00	38.0	$1.07\times$
FairSIN	57,621	4.40	39.1	$1.10\times$
Ours w/o FSER (confounded)	22,209	1.69	43.4	$1.22\times$
Ours w/o FSER (clean arm)	38,977	2.97	72.0	$2.03\times$
TrustFedGNN (ours)	38,979	2.97	73.2	$2.06\times$

7.8. How faithful is the contribution score to exact Shapley?

Not faithful enough to be called an approximation, and we tested this rather than assuming it. Against exact Shapley values over 125 probe points (Table 22), the pooled Pearson correlation is 0.7662 against its pre-registered threshold of 0.80, and pooled Spearman 0.7114, which clears its threshold of 0.70; sign agreement is 79.2% against an 85% threshold, bottom-1 detection is 68.0% against 75%, and mean simplex L_1 distance is 0.7453 against 0.15. With four of five criteria missing their targets, hypothesis H3 is refuted.

Table 22 reports the probe. We therefore do not claim that φ_k approximates the exact Shapley value. It functions as a fast, first-order ranking heuristic costing $O(KP)$ against $O(2^K P)$, and the primary contribution of the aggregation rule is the structural removal of the metadata dependence (Theorem 7), not fidelity to an exponential cooperative-game solution concept.

Table 22: **Empirical Evaluation of FU-Alignment vs Exact Shapley (125 probe points, $K = 5$, 5 seeds).** Evaluating across 5 probing rounds confirms FU-Alignment functions as a fast, first-order ranking heuristic ($O(KP)$ vs $O(2^K P)$) with strong directional alignment ($\rho = 0.711$, 79.2% sign agreement).

Metric	Target Criterion (Pre-reg)	Empirical Value (125 points)
Pooled Pearson Correlation r	≥ 0.80	0.7662 ($p < 0.001$)
Pooled Spearman Rank Correlation ρ	≥ 0.70	0.7114 ($p < 0.001$)
Sign Agreement Proportion	$\geq 85\%$	79.20%
Bottom-1 Detection Rate	$\geq 75\%$	68.00%
Mean Simplex L_1 Distance	≤ 0.15	0.7453
Computational Complexity	–	$O(KP)$ vs $O(2^K P)$

Table 23: **Composite Trust Score Sensitivity Analysis under Weight Perturbations (Issue I11).** Robustness of relative baseline rankings under 2,000 Monte Carlo weight configurations perturbing the five trust sub-score weights (Utility, Fairness, Privacy, Robustness, Calibration). Reports Spearman’s ρ_s , Kendall’s τ , and Rank-1 retention probability for TrustFedGNN across uniform $\pm 25\%$ perturbations and unconstrained Dirichlet(1) priors.

Perturbation Model	Spearman ρ_s (95% CI)	Kendall τ (95% CI)	Rank-1 Retention (%)
Uniform $\pm 25\%$ Perturbation	0.965 [0.764, 1.000]	0.923 [0.692, 1.000]	100.0%
Unconstrained Dirichlet(1) Prior	0.777 [0.341, 0.989]	0.677 [0.282, 0.949]	98.9%

7.9. Trust governance

The contribution decomposition of Proposition 10 gives the operator a per-client, per-round record of why each participant was weighted as it was, split into utility and fairness terms. Around it we report MC-dropout epistemic uncertainty [45], expected calibration error, and a composite trust score aggregating five axes.

Rank stability of that composite score was tested by Monte Carlo over 2,000 weight configurations, in two regimes, both of which we report. Under $\pm 25\%$ perturbation of the sub-score weights, Spearman $\rho_s = 0.965$ with 100% rank-1 retention; under an unconstrained Dirichlet(1) prior—which permits weight vectors no operator would choose— $\rho_s = 0.777$ with 98.9% retention. Quoting only the first regime would be selective reporting.

The operator-facing controls are τ and ϵ , each a first-class reportable quantity, and α is reported alongside them although a sweep over $\alpha \in [0, 1]$ produced no dose-response we could detect at ten seeds per setting (Section 7.5), which is what makes the system auditable against EU AI Act Articles 10, 13, 14 and 15 and the NIST AI RMF. Attention weights are

used here as behavioural diagnostics; we do not present them as sample-level causal explanations.

8. Limitations

Seven hypotheses were pre-registered prior to the empirical campaign and subsequently refuted or qualified by the measurements. Transparent reporting of null and refuting outcomes establishes the operational boundaries of the framework and adheres to rigorous empirical methodology.

1. **FSEER does not improve utility (P1).** $\Delta\text{AUC} = -0.0017$ at $p = 0.2324$ on Pokec-z, winning 3/10 seeds. Its measured value is as a safe inductive prior with a zero-distortion limit, not as a source of gain (Section 7.4).
2. **FSEER does not improve benign fairness.** On German, $\Delta\text{DPD}_{\text{hard}} = +0.0090$ at $p = 0.7344$: null on both axes in benign operation.
3. **We are not fairest, and on two of three settings we are matched or beaten (P3).** On Pokec-z our DPD_{hard} is second-worst of ten rows and our EOD worst. On Credit Default, FedAvg-GCN is nominally ahead on both axes, though both margins lie inside noise. Most starkly, on Bail across the full Dirichlet sweep TrustFedGNN trails plain FedAvg on AUC in *all twelve* settings of (α, K) , by 0.006 to 0.096, while its disparity is lower in only five of the twelve (Section 7.5). The utility result of Section 7.6 is therefore specific to Pokec-z and does not generalise across our own benchmarks. The paper’s defensible claims are the three guarantees—statistic-channel privacy, Byzantine resilience, and metadata independence—not superiority on group fairness or on utility in general.
4. **Defense components have narrow empirical support, and gradient clipping binds benign updates (P5, P6, T3).** Both the two-tier holdout (M6) and FairScore (M5) are null in benign operation. Under attack they separate. Removing the two-tier holdout significantly worsens disparity under fairness poisoning ($\Delta\text{DPD}_{\text{hard}} = +0.1019$, $p = 0.0020$) and triples the adversary’s weight, so that tier is supported—but only against that attack; under gradient scaling the measured effect is null and runs in the wrong direction ($p = 0.6406$). FairScore is null under both attacks ($p = 0.7422$ scaling, $p = 0.7695$ fairness poisoning) and in the wrong direction. We retain FairScore for

the formal decomposition it provides (Proposition 10) and state plainly that we have no empirical evidence it improves any measured outcome. Crucially, pre-registered testing of gradient clipping ($c = 10.0$, H_{T3}) refutes the hypothesis that clipping acts merely as an inert outlier shield on benign data: under this threshold, the clipping boundary actively binds $59.8\% \pm 4.6\%$ of benign client updates on German and $7.6\% \pm 5.6\%$ on Bail. Disproportionate clipping predominantly impacts clients exhibiting localized topological skew or sensitive feature concentration whose natural update norms exceed the population median. Rather than a static global threshold, future implementations should employ adaptive quantile-based clipping. EMA, by contrast, has clear positive evidence (Section 7.3).

5. **Fairness under attack is directionally right but inconclusive (P8').** Although disparity under fairness poisoning is 0.0665 against FedAvg's 0.1605 ($2.4\times$ reduction), at $n = 10$ this yields paired Wilcoxon $p = 0.3750$, which does not clear the pre-registered significance threshold ($\alpha = 0.05$). We strictly classify this outcome as inconclusive rather than confirmatory.
6. **The homophily hypothesis is not established (P10).** FSER benefit scaling with h_s gives $p = 0.0583$ with two of four points null.
7. **The contribution score is not a Shapley approximation (H3).** Pooled $r = 0.7662 < 0.80$ (Section 7.8).

Seven further boundaries are structural rather than hypotheses, and we list them so that none has to be inferred.

8. **The privacy guarantee is conditional and partial.** It is conditional on the model snapshot (Remark 4): θ_t was itself trained using $\mathbf{s}_k$, so the certificate bounds the privacy loss of the statistic sequence given the parameter trajectory, not that of the end-to-end pipeline. And it is partial: the joint release $(\tilde{\boldsymbol{\mu}}_k, g_k)$ is not DP, because the update channel carries no guarantee and retains a measured 0.645 attribute-inference AUC.
9. **The comparison with DP-SGD is not like-for-like.** The two mechanisms protect different channels; the $115\times$ figure is the cost of protecting each mechanism's own channel, not two prices for the same protection.

10. **Byzantine resilience requires the robust variant.** The rectifier alone is defeated by gradient scaling ($w_{\text{adv}} = 0.2008$). Only the median-screened variant holds $w_{\text{adv}} = 0.0000$ across all four scenarios, and it presumes a known or estimated corruption count $f < K/2$. Against the projected stealth adversary, which deliberately sits close to the benign median, the screen is not merely unhelpful but counterproductive (Section 7.3).
11. **Attack coverage and omniscient alignment adversary (T_1, H_{T_1}).** Pre-registered evaluation of an optimization-based alignment adversary (H_{T_1} , Adam optimizing client gradient direction to maximize alignment with g_{target} while injecting sensitive attribute bias, $n = 10$ seeds) refutes the hypothesis that cosine holdout gating alone resists omniscient white-box adversaries. The adversary successfully synchronizes its update direction with g_{target} , securing an adversarial weight share of $w_{\text{adv}} = 0.8514 \pm 0.0652$ on German Credit and $w_{\text{adv}} = 0.7461 \pm 0.0688$ on Bail Recidivism (both exceeding the 0.2000 rejection threshold, rejecting H_{T_1}). This formal negative outcome establishes the operational defense boundary of server reference gating: when an omniscient adversary has exact knowledge of g_{target} and optimizes directional alignment, holdout cosine filtering must be augmented with secondary norm and coordinate screening. We have not yet swept intensity downward toward the detectability boundary, nor tested distribution mismatch attacks on the server reference.
12. **The server holdout is stress-tested for size but not for representativeness.** Sweeping the requested holdout over $\{50, 100, 200, 400\}$ nodes moves no metric by more than two seed-level standard deviations, and the response is not monotone (Section 7.5); German caps the holdout at 125 nodes, so the claim is scoped to $[50, 125]$ and we do not extrapolate. What we did not vary is whether the holdout is representative of the client population, and a mis-specified target gradient remains the most plausible failure mode of this design.
13. **Absence of a validated centralised baseline anchor.** Centralised GNN training under comparable group fairness regularization undergoes optimization collapse to degenerate predictors ($\text{AUC} \approx 0.58\text{--}0.60$, Section 7.5), precluding its use as a meaningful ceiling anchor. Consequently, the exact quantitative decomposition of the federated utility gap into graph partitioning loss versus optimization dynamics remains an open empirical question.

14. **Determinism and scale (P5-X3, H_{P5-X3}).** GPU tables are not bit-exact reproducible, because aggregation on CUDA uses atomics. We bound the run-to-run dispersion directly rather than inferring it from across-seed variance: repeating the backbone-fixed control at identical seeds moves the paired contrast by 0.002 AUC (Section 7.4), which is about thirty times smaller than the margins in Table 18 and comparable to the magnitude of contrasts treated as null throughout this work. The two benchmarks held to the full ten-seed, seven-baseline protocol are mid-scale (67.8k and 30k nodes); the million-node run establishes computational feasibility only, having executed without the privacy mechanism. Crucially, on the large-scale Pokec-z social graph ($N = 67,796$, $K = 10$, $n = 10$ seeds, H_{P5-X3}), under Byzantine magnitude scaling ($c = 100$) TrustFedGNN demonstrates complete numerical stability and preserves uncompromised utility ($\text{AUC} = 0.7856 \pm 0.0122 \geq 0.7500$). However, elevated topological variance across large client partitions widens benign gradient dispersion, allowing the scaled adversary to retain $w_{\text{adv}} = 0.2984 \pm 0.0763 > 0.0500$, which formally refutes H_{P5-X3} on weight suppression and delineates graph scale boundaries.
15. **The fairness weight sweep is underpowered.** Sweeping $\alpha \in \{0, 0.05, 0.1, 0.25, 0.5, 1.0\}$ at $n = 10$ seeds is statistically underpowered ($d \approx 0.474$, $p = 0.1562$). While seed-level variance absorbs any monotonic trend at this cohort size, this leaves open the possibility of a weak dose-response that would require larger cohorts to resolve (Section 7.5).

9. Conclusion

Fairness-aware federated aggregation has been built, almost universally, on statistics that clients report about themselves. That design makes the fairness mechanism the softest target in the protocol: a participant that claims to be fair is rewarded for the claim. TrustFedGNN removes the dependence rather than hardening it. The server manufactures its own bi-objective reference gradient on a holdout split no client can reach, scores each participant by alignment with it, and gates those scores onto the simplex. Falsifying a report is then not merely detectable but inert—a client that lies leaves the weight vector bit-identical, while the same lie moves the weights of all six published rules we attacked, in 141 of 144 non-tied seed pairs.

Around that core, statistic-level differential privacy exploits the fact that the sensitive attribute enters the objective through exactly two scalars, buying a certified (ϵ, δ) release for $O(1)$ noise and 0.00175 AUC where full-parameter DP-SGD costs 0.20163. Four algebraic properties of the weight rule are machine-checked in Lean 4. We also prove that the repair one would reach for first—privatising the reported statistic—cannot work: once noise dominates, what the server observes converges to a constant of the noise itself, and the exact prediction matches measurement at every privacy level we tested.

We are equally clear about what the empirical evidence does not support. Edge-level structural regularisation (FSER) contributes nothing measurable to utility or benign fairness on homophilic or tabular graphs (Section 7.4). The contribution score is not a faithful Shapley approximation, and our measurements refute rather than assert that hypothesis (Section 7.8). Demographic disparity is not lowest-in-class on either flagship benchmark, and on Bail our utility trails unconstrained FedAvg across all twelve heterogeneity configurations (Section 8). When the GNN backbone is held fixed and only the aggregation rule varies, utility contribution on German is statistically neutral—confirming that the Pokec-z utility margin stems primarily from the architecture and normalization scaffold rather than the weighting mechanism. The differential privacy certificate conditions on the parameter trajectory and protects released demographic moments rather than parameter updates. Finally, while metadata independence is shared with FLTrust, FLTrust is fairness-blind by construction; TrustFedGNN achieves scaling-safe fairness steering through a bi-objective reference and payload norm-rescaling.

By addressing these foundational tensions directly, TrustFedGNN resolves the trilemma of algorithmic fairness, differential privacy, and Byzantine resilience without relying on unverified client disclosures. Rather than asking clients for disparity reports or corrupting constraints through noise, the coordinator evaluates reference signals on an independent validation holdout. Among surveyed federated graph learning frameworks, TrustFedGNN is the only architecture that both credits fairness improvements and is provably immune to metadata fabrication. This immunity is structural, certified algebraically in Lean 4, and validated down to bit-exact client weights. Coupled with payload norm-rescaling, TrustFedGNN provides a principled, scaling-safe foundation for trustworthy federated graph intelligence in regulated high-stakes domains.

Future work

The most consequential gap is theoretical, and we state it as an open problem for the community rather than as a limitation of this paper alone: *no convergence analysis exists for contribution-gated aggregation of this form*. The weight vector is recomputed every round from an inner product against a reference that itself moves with the global model, then passed through an EMA recursion and a rectifier that can zero coordinates outright—a non-expansive but discontinuous map. Standard federated-optimisation analyses assume weights that are fixed, or at least independent of the updates being weighted, and none of them covers this. Characterising the fixed points of the EMA recursion, and the conditions under which the gate stops switching, would put a class of contribution-based aggregators on the same footing that FedAvg has enjoyed for years. We have only the empirical behaviour of Table 17 to offer, and it is not a substitute. Closing the residual update channel calls for composition with secure aggregation or a full-update DP variant redesigned to tolerate FSER’s dependence on $\mathbf{s}$. On the empirical side, sweeping attack intensity downward toward the detectability boundary would test the defence where it is least characterised, and extending the ten-seed, seven-baseline protocol to the remaining benchmarks would broaden the utility claim. Finally, the fairness–homophily interaction suggests replacing the edge-level penalty with a mechanism that adapts to the measured h_s of the deployment graph, rather than assuming topological signal exists.

Appendix A. Formal Derivations and Proof Skeletons

Appendix A.1. Proof Skeleton for Theorem 3 (Minimax Observability Lower Bound)

This appendix provides the formal derivation and non-triviality verification for the minimax observability lower bound under local differential privacy.

Hypothesis Formulation.. Let $\widetilde{\mathbf{D}} = (\widetilde{\text{DPD}}_1, \dots, \widetilde{\text{DPD}}_K) \in \mathbb{R}_+^K$ denote the transcript of released disparity statistics observed by the server, where client k releases $\widetilde{\text{DPD}}_k = |\delta_k + \xi_k|$ with independent Gaussian noise $\xi_k \sim \mathcal{N}(0, \tilde{\sigma}^2)$. We formulate the verification of a global fairness budget $\tau > 0$ as a minimax

hypothesis testing problem between two candidate states:

$$H_0 : \delta_k = 0 \quad \forall k \in \{1, \dots, K\} \quad (\text{strict demographic parity holds everywhere}), \quad (\text{A.1})$$

$$H_1 : \delta_k = \tau \quad \forall k \in \{1, \dots, K\} \quad (\text{uniform boundary violation of the fairness budget}). \quad (\text{A.2})$$

Let $\mathcal{P}_0 = \{P_0\}$ and $\mathcal{P}_1 = \{P_1\}$ denote the probability distributions of the transcript $\mathbf{D}$ under H_0 and H_1 , respectively. The minimax decision risk for any test $\psi : \mathbb{R}_+^K \rightarrow \{0, 1\}$ is:

$$\mathcal{R}^* = \inf_{\psi} \max \left\{ \Pr_{H_0}[\psi = 1], \Pr_{H_1}[\psi = 0] \right\}. \quad (\text{A.3})$$

Le Cam's Two-Point Lemma.. By Le Cam's two-point lower bound [52, 53]:

$$\mathcal{R}^* \geq \frac{1}{2} (1 - \text{TV}(P_0, P_1)), \quad (\text{A.4})$$

where $\text{TV}(P_0, P_1) = \frac{1}{2} \int |p_0(\mathbf{x}) - p_1(\mathbf{x})| d\mathbf{x}$ is the total variation distance between P_0 and P_1 .

Data-Processing Inequality on Deterministic Folding.. Consider testing between composite hypotheses $H_0 : \text{DPD}_k \leq \tau - \gamma \quad \forall k$ and $H_1 : \text{DPD}_k \geq \tau + \gamma \quad \forall k$ with margin $\gamma \in (0, \tau)$. By Le Cam's two-point reduction, the minimax risk over composite hypotheses is bounded below by testing between any pair of least favorable configurations $\delta_0 \in H_0$ and $\delta_1 \in H_1$. Selecting boundary configurations $\delta_0 = (\tau - \gamma)\mathbf{1}$ and $\delta_1 = (\tau + \gamma)\mathbf{1}$ with canonical separation $\gamma = \tau/2$ yields coordinate-wise separation $\Delta\delta = 2\gamma = \tau$.

Let $\mathbf{Z} = (Z_1, \dots, Z_K)$ be the unperturbed, pre-folded Gaussian vector, where $Z_k \sim \mathcal{N}(\delta_k, \tilde{\sigma}^2)$. Under the selected configurations δ_0 and δ_1 , the distributions of $\mathbf{Z}$ are multivariate isotropic Gaussians:

$$P_{0,\mathbf{Z}} = \mathcal{N}\left((\tau/2)\mathbf{1}, \tilde{\sigma}^2 I_K\right) \quad \text{and} \quad P_{1,\mathbf{Z}} = \mathcal{N}\left((3\tau/2)\mathbf{1}, \tilde{\sigma}^2 I_K\right). \quad (\text{A.5})$$

The observation transcript $\widetilde{\mathbf{D}} = T(\mathbf{Z})$ is obtained by the coordinate-wise absolute value mapping $T(\mathbf{z}) = (|z_1|, \dots, |z_K|)$. Because T is a deterministic Markov kernel, the data-processing inequality for total variation distance guarantees:

$$\text{TV}(P_0, P_1) \leq \text{TV}(P_{0,\mathbf{Z}}, P_{1,\mathbf{Z}}). \quad (\text{A.6})$$

Pinsker's Inequality and KL Divergence.. By Pinsker's inequality:

$$\text{TV}(P_{0,\mathbf{z}}, P_{1,\mathbf{z}}) \leq \sqrt{\frac{1}{2} \text{KL}(P_{0,\mathbf{z}} \parallel P_{1,\mathbf{z}})}. \quad (\text{A.7})$$

The Kullback–Leibler divergence between two K -dimensional isotropic Gaussians with mean separation vector $\tau \mathbf{1}$ is:

$$\text{KL}(P_{0,\mathbf{z}} \parallel P_{1,\mathbf{z}}) = \frac{\|(3\tau/2)\mathbf{1} - (\tau/2)\mathbf{1}\|_2^2}{2\tilde{\sigma}^2} = \frac{\|\tau\mathbf{1}\|_2^2}{2\tilde{\sigma}^2} = \frac{K\tau^2}{2\tilde{\sigma}^2}. \quad (\text{A.8})$$

Substituting this into Pinsker's bound gives:

$$\text{TV}(P_0, P_1) \leq \sqrt{\frac{K\tau^2}{4\tilde{\sigma}^2}} = \frac{\tau\sqrt{K}}{2\tilde{\sigma}}. \quad (\text{A.9})$$

Combining with Le Cam's inequality in Eq. (A.4) yields the stated minimax observability lower bound:

$$\inf_{\psi} \max \left\{ \Pr_{H_0}[\psi = 1], \Pr_{H_1}[\psi = 0] \right\} \geq \frac{1}{2} \left(1 - \frac{\tau\sqrt{K}}{2\tilde{\sigma}} \right). \quad (\text{A.10})$$

Non-Triviality Verification at the Operating Point.. To demonstrate that this lower bound is informative rather than trivial (0), we evaluate it under the operational conditions of our cross-silo federation on German credit ($K = 10$, $n = 25$ nodes/group/client, target fairness budget $\tau = 0.05$):

- **Single-release regime** ($T = 1$): With $\epsilon = 8.0$ and $\delta = 10^{-5}$, the single-step standard deviation is $\tilde{\sigma} \approx 0.0484$. Here, $\tau\sqrt{K}/(2\tilde{\sigma}) \approx (0.05 \times 3.162)/(2 \times 0.0484) \approx 1.633 > 1$, which yields a trivial lower bound ≤ 0 . Testing is theoretically feasible in a single one-shot audit without repeated iterations.
- **Iterative composed regime** ($T = 60$ rounds): Accounted under Rényi differential privacy (Corollary 2), composing over $T = 60$ training rounds inflates the required per-round noise multiplier, yielding $\tilde{\sigma} = 0.3753$. At this operating parameter:

$$\frac{\tau\sqrt{K}}{2\tilde{\sigma}} = \frac{0.05 \times \sqrt{10}}{2 \times 0.3753} = \frac{0.1581}{0.7506} \approx 0.2106. \quad (\text{A.11})$$

The resulting minimax misclassification error is strictly lower-bounded by:

$$\mathcal{R}^* \geq \frac{1}{2} (1 - 0.2106) = 0.3947 \approx 39.5\%. \quad (\text{A.12})$$

Thus, under realistic repeated-round cross-silo training, any statistical rule evaluating whether the federation violates the fairness budget incurs an error of at least 39.5%—bounded away from zero, and within 10.5 points of the 50% random-guessing ceiling.

Declaration of Generative AI and AI-Assisted Technologies

During the preparation of this work the authors used AI-assisted tooling for code auditing, artifact verification and language editing. The authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

CRedit Authorship Contribution Statement

Ngoc-Son-An Nguyen: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Quang-Vinh Dang:** Conceptualization, Supervision, Funding acquisition, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and Code Availability

All datasets are publicly available benchmarks. Source code, test suites, result artifacts with provenance manifests, and the scripts that generate every table in this paper are available at <https://github.com/vinhqdang/FedFairGNN>.

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